Rehman *et al. Journal of Big Data* (2025) 12:264
https://doi.org/10.1186/s40537-025-01323-2

Journal of Big Data

SURVEY Open Access

A systematic literature study of machine learning techniques based intrusion detection: datasets, models, challenges, and future directions

Hafiz Muhammad Raza Ur Rehman[1†], Saira Liaquat[2†], Muhammad Junaid Gul[1], Muhammad Zeeshan Jhandir[2*], Daniel Gavilanes[3,4,5], Manuel Masias Vergara[3,6,7] and Imran Ashraf[8*]

[†]Hafiz Muhammad Raza Ur Rehman, Saira Liaquatt have contributed equally to this work.

*Correspondence:
Muhammad Zeeshan Jhandir
zeeshan.jhandir@iub.edu.pk
Imran Ashraf
imranashraf@ynu.ac.kr

Full list of author information is available at the end of the article

Abstract
The tremendous growth of cyberspace has magnified the risk of advanced cyber-attacks, making the conventional security system deficient. An intrusion detection system (IDS) is a crucial aspect for securing the networks, and machine learning based approaches have attained significance for detecting growing threats, which is extremely essential to address the major concern of network security. This survey paper systematically reviews the machine learning-based IDS, focusing on detection models, the most used datasets, and evaluation metrics. A systematic review methodology, including defined selection criteria and a detailed analysis framework, enables clarity and reproducibility. Furthermore, the paper presents a comparative analysis of popular IDS datasets such as NSL-KDD, CICIS2017, and UNSW-NB15 and points out their limitations, including imbalance and antiquated attack methods. The key contributions are: (a) an organized taxonomy of ML-based IDS techniques, (b) a detailed comparison of datasets and related challenges, and (c) pointing out the research gaps to guide the future directions. This survey paper aims to help scholars in developing vigorous, intelligent, and convertible intrusion detection solutions.

Keywords Network intrusion detection, Cyber security, Machine learning, Network security, Denial of service

Introduction

In the last few decades, internet technologies have advanced significantly and shown tremendous expansion in both online services and information sharing. In fact, the Internet evolved into a widely utilized public medium for communication and information distribution. Therefore, above 48% of people on the planet utilized the Internet as the primary source of information, and that number rose to 81% in developed countries [1, 2]. As a consequence of this growth, hackers and other invaders are using the internet as a target, which leads to an increase in cyber attacks [3]. It includes attempting to disturb

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

the normal behavior of a network through overloading machines, improper usage of software, and unauthorized access to credentials and files. Malicious activities, malware, ransomware, denial of service (DoS) [4], cryptographic attacks, zero-day attacks [5], illegal access, breaches of data, and phishing attacks are common attacks for the Internet of Things (IoT) devices. These kinds of cyber crimes and security problems have the ability to either interrupt the working operations or cause economic damage to both individuals and enterprises.

According to the IBM report-2023, a data breach affected the United States by about 8.19 million USD [6], and the overall annual losses of cybercrime to the worldwide economy are approximately 4.45 million USD [7]. The number of cyber crimes has skyrocketed, and that becomes a concerning threat to cyber security professionals and research scholars [8]. Therefore, to ensure the general national security of a nation's businesses, government, and individuals, security management systems that can identify and prevent such occurrences in a timely and intelligent manner are desperately needed. A more reliable and secure computer system must guarantee the privacy, accessibility, and accuracy of data. When an unlawful discriminating person, program, or other unauthorized entity enters a computer network with the intention of harming or interrupting the normal functions, the trustworthiness and safety of the system become compromised [9]. Cybersecurity involves a collection of techniques and security measures that have been used for protection against unauthorized access and attacks on user funds, as well as the cyberspace itself. The primary motive of the cyber defense system is to guarantee the accessibility, essentials, and security of data. Controls have been integrated with computer networks to ensure that authorized individuals can access data only.

The integrity of every nation is significantly influenced by its national defense. In January 2008, President Bush's government launched the 'Comprehensive National Cyber Security Initiative' (CNCSI) [10]. This initiative's primary objectives consist of bringing focus on a variety of concerns, such as the identification of emerging and present cybersecurity risks, the discovery and patching of historical cyber vulnerabilities, and the identification of those attempting to obtain access to confidential federal data. The next president of the US, Obama, enlarged this initiative and declared that "cyber-threat is one of the extremely serious economic and national security challenges we face as a nation and that 'America's economic development in the 21st century will depend on cyber-security" [10]. Intrusion detection systems (IDS) are network security tools that block intrusions, protect programs/data, and prevent illegal access to computer systems. The primary objectives of IDS are to promptly identify and understand a broad spectrum of intrusions, adapt to unfamiliar attacks, and recognize and comprehend intrusions promptly. A recent increase in cyber threats emphasizes the need to protect network resources. IDS makes it feasible to find, recognize, and understand unusual behavior caused by hackers in computer networks or systems. A cyber-attack is the kind of illegal attempt where a variety of attacking techniques have been utilized by any individual or organization to attain unauthorized access to a computer system from anywhere to harm the computer systems, including modifying, blocking, or stealing data.

Machine learning sub-domain deep learning models are applied to analyze data for various aspects (like detecting patterns, predicting trends, and identifying anomalies in real time). For example, traffic detection of a particular roadway [11]. It is also used for collecting data across a range of fields in order to improve and automate the process.

However, the majority of ML models are utilized for data classification and regression. The use of ML across various cybersecurity applications is growing exponentially. Even though an extensive amount of research on IoT attack detection using machine learning techniques is constantly being conducted, not much research exists about effective detection approaches [12].

There are five sub-categories of IDS as shown in Fig. 1. These categories are described as follows:

(i) Network IDS: It monitors the entire network's traffic and notifies whenever it detects any suspicion in the flow of traffic.
(ii) Host IDS: Thoroughly monitor the incoming and outgoing network packets for any suspicious activity through continuous tracking [13].
(iii) Protocol-based IDS: It tracks the development of a protocol (HTTP-Protocol) in order to assure the security of web-servers.
(iv) Application-protocol IDS: It monitors the connections among the applications by residing inside a server.
(v) Hybrid-IDS: It consists of the combination of a few aforementioned IDS that will increase attack detection [14].

The research related to ML-based IDS is still in its infancy in comparison to other ML-based applications; therefore, it deserves more attention than any other domain. It becomes easier to identify attacks and halt anomalous IoT behavior through utilizing ML models [15]. IoT data is constantly being generated in enormous amounts and remains stored on big data servers that are accessible through the internet. Nowadays, hackers have become more cunning and inventive since they have a variety of methods available for utilizing vast amounts of data. They knew well how to avoid firewalls or virus scans and confuse people. Large enterprises such as Adobe, AOL, and eBay have also experienced numerous cyberattacks. Regardless of having varying degrees of security in place to protect their data, these companies continued to suffer attacks. Intruders aim to target both small and large user types. Therefore, some sort of protocol and mechanism should be in place that protects general users from every sort of cyber-attacks [16–18].

The major objective of this research is to explore the major ML-based models that are used in cybersecurity, identify the current trend of ML approaches in cybersecurity, and identify suitable future work directions related to this domain. This domain is still fragmented and facing a deficiency in the synthesis of methodology, even with the combined efforts of scholars. It is now crucial to close this gap by conducting an extensive, systematic literature review that discusses intrusion detection in computer networks along with prior research, datasets, and evaluation methodologies, and clarifies

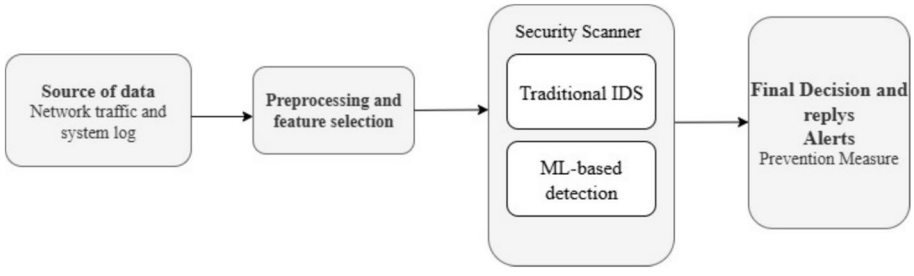

Fig. 1 A generic architecture of an intrusion detection system integrating with ML-based approaches

the ethical consequences. This study intends to provide the efficacy of cutting-edge ML techniques and how they have been or could be used to identify and categorize cyber-attacks, including intrusion detection on computer networks and mobile-based systems. As internet usage continues to rise, numerous security concerns have also arisen.

Institutions and organizations that work in this field are spending more money all the time to give their customers more reliable and secure services. Machine learning and network behavior analysis are two techniques used to identify attacks early on with the help of current technology. Therefore, intrusion detection has emerged as the most recent area of study for organizations in the literature and for cybersecurity-related organizations.

This survey aims to address gaps in the existing literature by systematically reviewing ML-based intrusion detection systems. The key contributions of this work are as follows;

i. **Systematic Review using PRISMA**: We follow a PRISMA-based methodology to ensure clarity and reproducibility, which is rarely applied in IDS surveys.
ii. **Comprehensive Dataset Analysis**: We compiled and provided a comparative analysis of major IDS datasets, identifying their limitations, challenges, and discussing their impact on IDS efficiency
iii. **Evaluation Metrics**: We integrate state-of-the-art results from recent IDS papers into a combined comparison of algorithms, datasets, and evaluation metrics.
iv. **IDS Categorizing and Tendency**: We classify the ML-based IDS techniques (supervised, unsupervised, deep learning, and hybrid) and pinpoint the latest research directions, such as sustaining stable performance, explainable artificial intelligence (XAI), and multimodal intrusion detection system [19].

The current study is different from existing surveys in various aspects, e.g., covering those aspects of IDS that are not or partially covered by existing works. A comparative overview of these aspects is provided in Table 1.

Figure 2 presents the overview of the current work and the contents it covers. For the rest of this review, the methodology is explained in Sect. "Methodology". Section "Cyber Security Systems" provides an overview of traditional and ML approaches for cybersecurity. Evaluation metrics for IDS are covered in Sect. "Evaluation Metrics for IDS". Section 12 is about the recent challenges encountered by the ML-based systems for cybersecurity. Future research directions are provided in Sect. "Future Work". Finally, the conclusion is given in Sect. "Conclusion".

Methodology

Figure 3 shows the process of finding and sorting the research articles for this review. The systematic review complied with the structured methodology for conducting a review mentioned by Kitchenham and his fellows [42], which consists of a series of different phases necessary for systematic planning, computing, and reporting. The process starts with developing relevant research questions to guide the study. Afterwards, a detailed list of keywords was developed to facilitate a thorough review of existing research using a well-defined approach to capture the most relevant research articles. This was observed by defining clear inclusion and exclusion standards to ensure a systematic and reproducible filtering process. In the next process, we answer the research questions, ensuring the relation and systematic organization of the survey. Lastly, the process concluded and

Table 1 Comparison of the current study with previous surveys (legend: ✓ indicates covered; ≈ indicates somewhat covered; ✗indicates not covered)

Ref.	Year	CIT	Cyber Security			Machine Learning					
			Mobile Based IDS	Computer Network IDS	Security Datasets	Techniques	Metrics	Tools	Trust	Adversarial ML	Graphical Summary
[20]	2016	31	✗	✓	✗	✗	✗	✗	✗	✗	✗
[21]	2016	501	✗	✓	✗	✗	✗	✗	✗	✗	✗
[22]	2017	55	✗	✓	≈	✗	✓	✗	✗	✗	✗
[23]	2017	325	✓	✓	✓	✓	✓	✗	✗	✗	✗
[24]	2018	1046	✗	✓	✓	✓	✓	✗	✗	✗	✗
[25]	2018	227	✗	✓	✗	✓	✗	✗	✗	✓	✗
[26]	2019	509	✗	✗	≈	✗	✗	✗	✗	✗	✗
[27]	2018	320	✗	≈	✗	✓	✗	✗	✗	✗	✗
[28]	2018	32	✗	≈	✗	✓	✗	✗	✗	✗	✗
[29]	2018	34	✗	✓	✗	✗	✗	✗	✗	✗	✗
[30]	2018	125	✗	✗	≈	✓	≈	≈	✗	✗	✗
[31]	2019	17	✗	≈	✗	✗	✗	✗	✗	✗	✗
[32]	2019	775	✗	✓	✓	✓	✗	✗	✗	✗	✗
[33]	2019	506	✗	✓	✓	✓	✓	✗	✗	✗	✗
[34]	2019	152	✗	✓	✗	✗	✓	✓	✗	✗	✗
[35]	2021	123	✗	✓	✗	✗	✗	✗	≈	✓	✗
[36]	2020	76	✗	✓	✓	≈	✓	✗	✗	✓	✗
[37]	2023	38	✗	✗	✗	✓	✓	✗	✗	✗	✗
[38]	2023	30	✗	✗	✓	✓	✓	✓	✗	✗	✗
[39]	2024	3	✗	✓	✓	✓	✓	✗	✗	✗	✗
[40]	2024	2	✗	✓	✓	✗	✗	✗	✗	✓	✗
[41]	2024	3	✗	✓	✓	✓	✓	✓	✗	✗	✗
Our	2024	–	≈	✓	✓	✓	≈	✗	✓	✓	✓

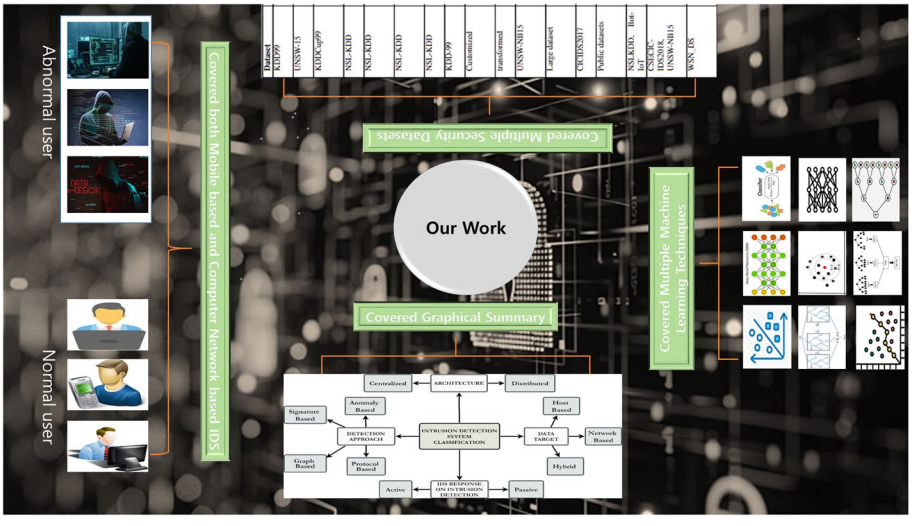

Fig. 2 Overview of the current work

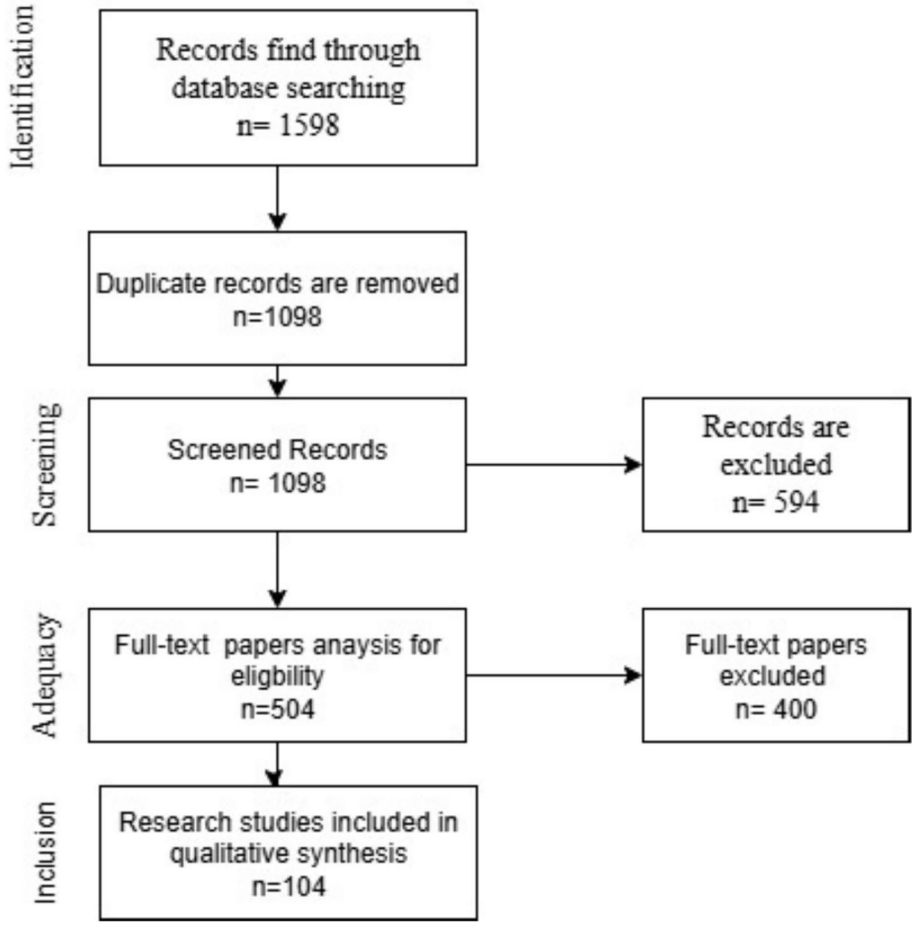

Fig. 3 PRISMA flow diagram for the systematic review

provided research synthesis about the information extracted, which was analyzed and integrated to provide the unified view of the study area, key design patterns, gaps, and challenges in ML-based intrusion detection of computer networks and mobile devices. In order to find pertinent work on machine learning and cyber security, searches were conducted through IEEE Xplore,[1] ACM,[2] digital library, Emerald Insight, Springer[3] Link, and Science Direct[4] for articles that contained the terms ("machine learning" and "cyber security"), ("machine learning" and "intrusion detection"), or both in the title, abstract, or keywords. Additionally, queries were made to Web-of-Science, Google-Scholar,[5] Scopus, and Semantic Scholar[6] in order to confirm the results and locate additional pertinent papers in lesser-known resources.

Research questions

The basic aim of this systematic review is to examine ML-based techniques with regard to IDS. The detailed analysis involved multiple ML-based models and techniques

[1] https://ieeexplore.ieee.org/Xplore/home.jsp [April 20, 2024 11:34 am]

[2] https://www.acm.org/ [April6,2024,2:00pm]

[3] https://link.springer.com/ [April14,2024,10:34am]

[4] https://www.sciencedirect.com/ [March28,2024,11:am]

[5] https://scholar.google.com/ [April18,2024,9:12am]

[6] https://www.semanticscholar.org/me/research [April 16,2024,2:45am]

developed particularly to tackle intrusion detection attacks in computer networks and mobile devices, covering the last ten years. The research questions are formulated to address the objectives systematically as follows:

- **RQ1:** What is the most recent state-of-the-art ML-based IDS in computer networks and mobile devices?
  This question determines state-of-the-art ML-based IDS applications and the most commonly used techniques and models in the context of computer networks.
- **RQ2:** Which datasets are frequently used for practical experimental analysis with ML-based IDS in computer networks and mobile devices?
  This question identifies the datasets used to practically evaluate ML-based models for IDS. Its main objective is to detect the data types that are useful in developing these models, their importance, and their real-time application in detecting intrusions.
- **RQ3:** What are the current limitations and future research directions in ML-based IDS in computer networks and mobile devices?
  This question highlights the shortcomings in current research and proposes areas for future exploration regarding the use and applications of ML in the context of computer networks and mobile devices.

Research strategy

This survey paper uses automated and manual search procedures to compile a detailed list of research articles aligned with the study's objectives. Automated search techniques were applied to databases such as IEEE Xplore, ACM Digital Library, Emerald Insight, SpringerLink, ScienceDirect, and Web of Science to ensure wide-ranging coverage of the literature. A manual search was also conducted to explore search engines and review the reference lists of related publications to identify the most relevant papers. The following structure of the search process was adopted in this systematic review:

- We selected short terms based on a comprehensive review of related literature to ensure both relevance and specificity in the search procedure.
- Afterwards, we broadened the search criteria to include variations and related terms. For example, we included "Intrusion Detection System" along with its acronym (IDS) to cover a detailed array of relevant papers and improve completeness.
- Example search string:
  ('Machine Learning and Cybersecurity" OR "Machine Learning Based" OR ML OR ML-based) AND ('Computer Networks, Mobile Devices" OR CNs OR "Intrusion Detection System" OR IDS).

To ensure comprehensive coverage, searches were conducted through IEEE Xplore, ACM, Emerald Insight, SpringerLink, ScienceDirect, Web of Science, Google Scholar, Scopus, and Semantic Scholar. Titles and abstracts from the first ten pages of search results (covering publications from 2011 to 2024) were carefully examined to confirm that no prior studies directly overlapped with our structured survey.

Inclusion and exclusion criteria

In this stage, the most relevant papers were manually examined by three authors independently. At the beginning, the researchers assessed titles, abstracts, and, when

necessary, the full texts to examine their relevance. The following criteria were used to include a research article:

- **Scope of Publication**: Only original research published in peer-reviewed journals was included to ensure quality and depth.
- **Relevance to IDS**: The research must specifically address IDS and its related domain.
- **Use of ML-based techniques**: Only studies applying ML-based techniques to IDS in computer networks or mobile devices were considered.
- **Publication Period**: Studies published between 2011 and 2024 were included to capture recent advancements.

To exclude an article as being unsuitable for this review, the following criteria were used.

- **Unrelated Concept**: Studies mentioning ML without applying it to IDS, or mentioning IDS without ML, were excluded.
- **Publication Type**: Conference papers, proposals, secondary sources (surveys, reviews), workshops, and letters were excluded. Only primary research with novel contributions was considered.
- **Language**: Papers not written in proper English were excluded.

Article selection

This section explains the process of selecting research articles and filtering out irrelevant ones. Figure 3 shows the PRISMA flow diagram for our systematic review.

The initial retrieval identified 1598 papers from different databases (IEEE Xplore, Google Scholar, ACM, etc.). After removing duplicates, 1098 papers remained. Out of these, 594 papers were checked for eligibility based on predefined inclusion and exclusion criteria (e.g., ML-based IDS in computer networks, publication year, peer-reviewed status). Papers without sufficient methodological details or unrelated focus were excluded. After careful assessment, 104 papers were included in the final analysis and subjected to an in-depth review.

Analysis of included papers by publication year

Figure 4 illustrates the frequency of ML-based IDS research contributions across different countries. The data show that scholars from India, the USA, China, and others contributed the most, indicating strong global research interest in applying ML to secure computer networks and mobile devices between 2011 and 2024.

The list of acronyms that have been used in this article is given in Table 2.

Figure 4 shows the distribution of the chosen articles for this study, showing that more articles were published in the last few years, with 2018 and 2019 having the highest number of posts. Because the present selection only covers the period up to April 2024, some of the articles from 2025 are also added, but a larger number of articles are anticipated for 2023 and 2024.

Cyber security systems

An act that poses a serious danger to privacy, the availability of necessary, valuable information, and systems is referred to as a security breach. Numerous security lapses compromise the privacy of a network, an organization, or a single user. Below is a discussion of a few of them.

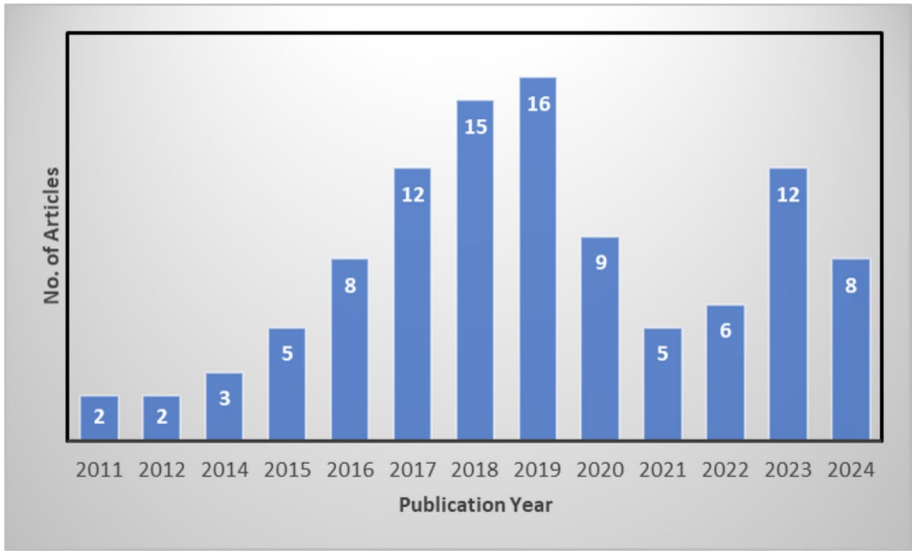

Fig. 4 Number of articles published in each year

Table 2 List of acronyms

Acronyms	Description
AI	Artificial Intelligence
AUC	Area under Curve
CNCSI	Comprehensive National Cyber Security Initiative
DoS	Denial of Service
FP	False-Positive
FN	False-Negative
IDS	Intrusion Detection System
ID	Intrusion Detection
IoT	Internet-of-Things
IP	Internet Protocol
K-NN	K-Nearest Neighbor
LR	Linear-Regression
ML	Machine-learning
MLP	Multi-layer Perceptron
NB	Naive Bayes
RF	Random Forest
R2L	Remote to Local
U2R	User to Root
WSN	Wireless Sensor Network

i. Unauthorized entry into data from a system, network, or data resource is referred to as illegal access.
ii. A software that damages a computer, server, client, or computer network is known as malware. Computer viruses, worms, spyware, adware, and malicious bots are just a few examples of the various types of malware, which are harmful software. [16].
iii. A cyber-attack in which an unauthorized user attempts to stop a computer or other device from performing its intended functions by preventing it from doing so is called a denial of service (DoS) attack [4]; It causes a DoS for additional users by isolating the targeted system with requests until regular traffic cannot be handled.

iv. Phishing is the false effort to transmit emails from a reliable source to a targeted system. According to [43], the communication seems to be from a reliable user. This is the most prevalent type of cyberattack, and its primary goal is to steal confidential data. Proper education and training of the user is one approach to rescue ourselves.
v. A day without an attack occurs when a hacker exploits a software flaw before the developer has a chance to identify and address it [44]. It can manifest itself in a variety of ways, including flaws, SQL injection, flawed algorithms, URL redirection, buffer overflows, and missing data encryption.

Table 3 provides a brief overview of cyber attacks and classes of attacks on different devices. Privilege rises, password assaults, insider threats, advanced persistent threats [16], man-in-the-middle attacks, web application attacks, crypto-currency attacks, and SQL injection attacks are some of the additional security flaws and problems. Protection tools like firewalls, IDS, and antivirus software make up any network security system. Any unauthorized infiltration with malicious intent may be found and identified with the use of IDS. There are three primary categories of network analysis for IDS:

(a) Signature-based, which ignores a high false alarm rate (FAR) to identify known assaults
(b) Anomaly-based, which looks for unusual system and network activity,

Table 3 Intrusion attacks on personal computers, servers, and mobile devices

Type of intrusion attack	Classes of attack	Description
Attacks on personal computers		
Malware (e.g., Viruses, Trojans, Ransomware)	HIDS	Software intended to steal data, cause computer damage or disruption, or get unauthorized access.
Phishing	Social Engineering attacks	Deceptive attempts to pose as a reliable organization in order to gain private information.
Keylogging	Host-based attack	Capturing keystrokes to gather sensitive information such as passwords and personal data.
Remote Access Trojans (RATs)	Host-based Malware attack	, which allows an attacker to control a computer remotely.
Denial of Service (DoS)	Injection attack	Attacks meant to prevent intended users from accessing a computer resource.
Attacks on servers		
SQL Injection	code Injection Attack	Inserting malicious SQL code into queries to manipulate or access the database.
Cross-Site Scripting (XSS)	Password and Authentication attack	Injecting malicious scripts into web pages viewed by other users.
Brute Force Attacks	Network-based Attacks	Repeatedly trying various passwords to gain unauthorized access to a server.
Distributed Denial of Service (DDoS)	Traffic flooding attack	Overwhelming a server with requests from many sources in order to interfere with services.
Attacks on mobile devices		
Mobile Malware (e.g., Spyware, Adware)	Malware-based attacks	Malicious software specifically targeting mobile operating systems like Android and iOS.
SMS Phishing (Smishing)	Social Engineering attack	Phishing attacks conducted through SMS messages to steal personal information.
SIM Swapping	Identity-Theft attack	Gaining control of a mobile phone number by tricking the carrier into transferring it to a new SIM card.
Malicious Apps	Malware-based attack	Applications that appear legitimate but perform harmful activities once installed.

(c) Hybrid-based, reduces FAR for unknown threats by combining (a) and (b). Probing is an attack in which a cybercriminal searches a computer system or network to find vulnerabilities and weaknesses that can be used in the future.

Due to the ease with which a slight alteration in the attack pattern might circumvent the signature-based IDS, ML approaches outperformed traditional signature-based systems. Systems that rely on ML, however, pick up knowledge from traffic patterns. The assault versions are clearly detectable by them. On the other hand, ML-based systems pick up knowledge from traffic patterns. They can quickly identify the many types of attacks. Furthermore, because ML-based systems do not analyze every signature in the database, their CPU loads range from low to moderate. Additionally, ML-based systems function faster and more accurately while obtaining and revealing the intricate characteristics of attack activity. Although other taxonomies and attack classifications have been presented by the researchers, intrusion detection, categorization, and detection are the primary topics of this article.

Cybersecurity offers numerous benefits; however, it also has drawbacks, such as the potential for harm from competitors and criminals. Because of the ongoing participation of criminals, there are more intrusion occurrences occurring every year [45]. A network security tool called an IDS keeps an eye out for and identifies unauthorized or unlawful activities occurring within a computer system. An IDS is evaluated based on three criteria: accuracy of attack detection (False +ve), completeness in identifying all threats (False -ve), and speed in detecting threats.

Table 4 provides a summary of cyber attacks on IoT devices, network devices, and industrial control systems [46], while intrusion attacks on medical devices are listed in Table 5. Phishing still remains one of the most common and developing cyber threats, now improved by artificial intelligence and digital technologies. Conversional detection methods have given way to advanced machine learning and deep learning approaches. Current research studies have explained the efficacy of deep learning in phishing email and website detection. For instance, a systematic review published in 2024 provides a detailed analysis of deep learning techniques for phishing detection and their linked challenges [47]. The software and tools that are used to detect attacks and the ensemble model achieved a 96 percent detection accuracy on standard datasets [48]. Although Phish Guard, a multi-layered ensemble system, reported nearly 99 percent accuracy in detecting phishing websites [49]. In addition, reinforcement learning-based models, such as DQN-based adaptive phishing detection, have shown great performance in a dynamic attack framework [50]. Moreover, the enhancement of AI-driven phishing awareness workshops using generative models and social engineering tactics has resulted in a significantly increased attack complication, making new detection systems captious [51, 52].

In their explanation of IDS tools employing ML models, Maxwell et al. [53, 54], point out that creating precise and effective ML models requires more than just raw network traffic data. It is necessary to convert this data into structured vectors instead. To adequately capture the data in the model, the feature engineering process necessitates a combination of ML approaches and domain expertise. The network architecture, networking type, data gathering, and other crucial elements are depicted in the system implementation in the following figure. While timeliness offers information on

Table 4 Intrusion attacks on IoT devices, network devices, and industrial control systems

Type of intrusion attack	Class of attack	Description
Attacks on IoT devices		
Botnets	Network-based Attacks	Networks of infected IoT devices used to carry out attacks such as DDoS.
Firmware Exploits	Hardware or low-level attack	Attacks that exploit vulnerabilities in the device's firmware.
Unauthorized Access	Access control Attack	Gaining control over IoT devices to manipulate their operations or gather data.
Man-in-the-Middle (MitM) Attacks	Network-based attacks	Modifying and intercepting two devices' communication.
Attacks on network devices		
Router Attacks	Network-based attacks	Exploiting vulnerabilities in routers to intercept, alter, or reroute traffic.
Man-in-the-Middle (MitM) Attacks	Network-based attacks	Eavesdropping on or changing communication among network devices.
DNS Spoofing	Network-based attack	Redirecting traffic to malicious sites by altering DNS records.
Packet Sniffing	network-based attack	Capturing and analyzing network packets to steal information.
Attacks on industrial control systems		
SCADA Attacks	Critical Infrastructure	Targeting data collecting and supervisory control systems to disrupt industrial operations.
PLC (Programmable Logic Controller) Attacks	Industrial Control System (ICS)	Manipulating PLCs to cause malfunctions in industrial processes.
Stuxnet-like Malware	Advanced Persistent Threat	Highly sophisticated malware designed to target specific industrial control systems.
Sabotage	Cyber-Physical Attacks	Physical or cyber attacks aimed at damaging industrial infrastructure.

Table 5 Intrusion attacks on medical devices

Type of intrusion attack	Class of attack	Description
Ransomware	Data Exfiltration or Information Disclosure Attack	Demanding a payment to unlock vital medical data that has been encrypted.
Data Breaches	Cyber-Physical Attacks	Stealing sensitive medical information from devices.
Device Tampering	Access Control Attack	Physically or remotely altering the functioning of medical devices.

granularity, detection response, and detection timing, the detection strategy addresses the processing approach, discipline, and methodology.

Figure 5 depicts the overview of ML-based IDS. ML is a subfield of artificial intelligence (AI), which is concerned with creating systems that can replicate human cognitive processes. Machine learning, which may be further classified as supervised, unsupervised, or reinforcement learning, allows machines to solve complicated problems in a manner similar to that of humans. Large-scale data training enables ML models to predict new data accurately without the need for explicit programming. In the subject of cyber-security, which is becoming more and more hampered by sophisticated cyber-threats, this technology has important implications. The efficacy of ML techniques in tackling cybersecurity issues in a variety of fields is demonstrated by many research works that demonstrate how they may help identify and respond to cyber attacks.

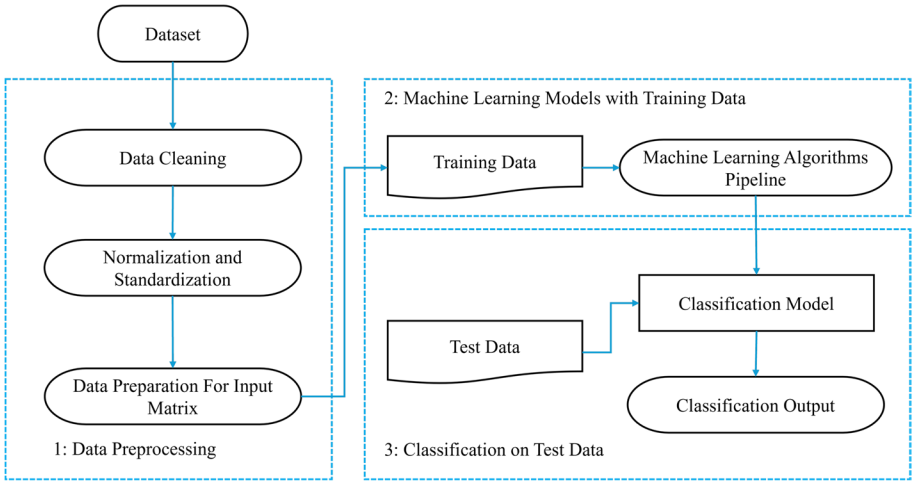

Fig. 5 Overview of ML-based IDS system

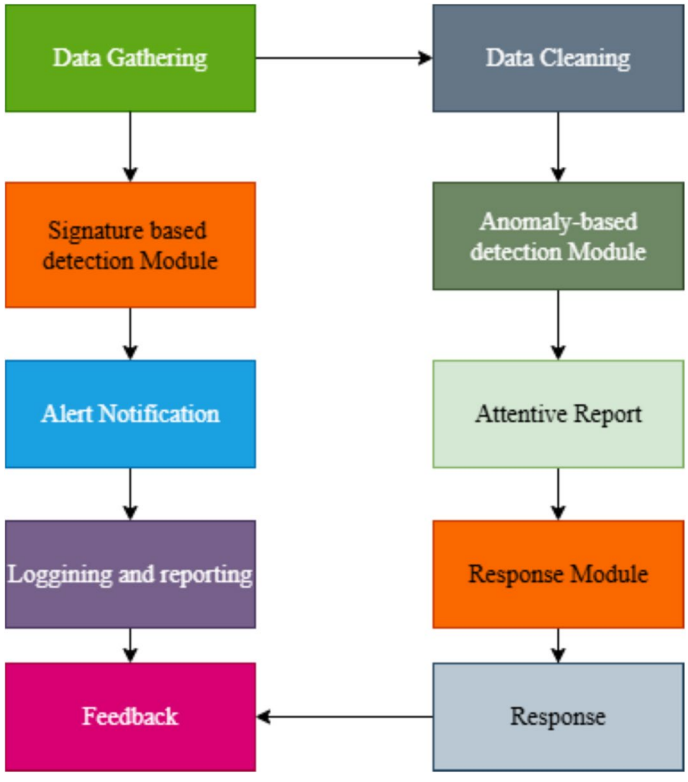

Fig. 6 An overview of a traditional intrusion detection system [55]

Traditional intrusion detection system

Figure 6 depicts a high-level synopsis of IDS. In data collection, the unprocessed data comes from various sources and is gathered for analysis. Raw data is used for preprocessing to clean, filter, and normalize for effective analysis. In contrast to the signature-based detection module, which analyzes data patterns against the database of familiar attack signatures, the anomaly-based detection module establishes a baseline of typical behavior and marks departures from it as potentially odd. In the alert notification, the IDS generates alerts when suspicious activities are identified, indicating a potential

intrusion. Alerts are an indication and are communicated to administrators or security personnel for further investigation. In logging and reporting, detailed logs and reports are generated for post-analysis. The configuration and severity, depending on the IDS, may activate predefined replies, such as blocking IP addresses or changing access controls. Feedback from the responses is used to enhance the IDS's signature database or abnormality detection parameters, improving its accuracy over time. This block diagram shows a high-level overview of the processes involved in a conventional IDS model. It runs down the key stages from data collection to response, and highlights the coupled nature of anomaly-based and signature-based detection schemes.

Network-based IDS using ML models

IDSs that are based on anomaly have been introduced in order to combat the new attackers' explosive expansion. This sort analyzes network traffic and determines if it is malicious or benign by using ML from the data [56]. Anomaly-based IDS may identify known and new threats without requiring either human intervention or explicit programming. The ML methods most frequently employed to implement NIDS are reviewed in the next subsections. Classification is the process of making sets of data items among several classes. Furthermore, binary classification—in which the input can only have two values—or multi-class classification—in which the output can have more than two values—are two possible classification techniques.

Table 6 shows a list of different ML approaches for intrusion detection along with commonly used security datasets. According to [24, 57], ML models are considered supervised learning, where classifiers learn from labeled datasets before making predictions based on fresh data. [58] conducted a study which is a network-based intrusion detection system and highlight the previously used datasets that influence the performance of the IDS algorithm. The study [59] provides an extensive survey on IDS mainly focus on old and one of the updated datasets, machine learning and deep learning algorithms and also discuss key challenges facing by the real-time world community. The research [60] demonstrates the framework for detecting the cyberattacks using traditional classification algorithms (DT, RF, Naive Bayes, KNN, SVM), and IDS old datasets and practical application on it. Similarly, [61] conducted a survey based on machine learning

Table 6 Comparison of ML models' performance according to various datasets

Ref.	ML model	Datasets	# of attacks	Attributes	# of attributes	Accuracy (%)
[56]	Auto-encoder	KDDCup99	22	Features	41	95%
[71]	DNN	NSL-KDD	22	Sample	50,000	99.73%
[72]	KNN, DNN	DARPA, IDS	38	–	–	75.00%
[73]	XGBoost	UNSW-NB-15	9	Features	49	92%
[74]	RF, GBT	CICIDS2017	15	Features	83	–
[75]	SVM	NSL-KDD	–	–	–	92.00%
[76]	SVM, PCA	ISCX	–	–	–	–
[77]	XGBoost	NSL-KDD	–	–	–	N/A
[78]	SVM, LR	ISOT	–	Flows	1,675,442	–
[79]	SVM	ADFA Linux	7	Traces	5206	–
[80]	Clustering	CTU-13	–	Scenario	13	–
[81]	Hybrid ML	BOT-IOT	8	Features	49	99.90%
[82]	GA, LR	NSL-KDD	41	Feature	125,973	92.12%
[83]	ANN	HTTP CSIC-2010	3	HTTP Requests	61,000	–
[84]	Spark-Chi-SVM	KDD CUP-99	41	Features	494,021	96.80%

and deep learning solutions for intrusion detection and prevention on IoT environments and discussed the IoT-specific challenges and datasets. The authors in [62] present a survey based on deep learning models specifically used in IoT network intrusion detection and compare these DL-based models in IoT-related datasets. [63] conducted a study that mainly focuses on Machine learning techniques (DT, KNN, RF, ANN), applications in industrial control systems, and also highlights the lack of real-time ICS datasets. The survey [64] evaluates the machine learning techniques for intrusion detection in general-purpose computer networks, reviews commonly used ML models, and analyzes the strengths and applications of detecting DoS and Prob attacks. [65] conducted a study present comparative analysis on IDS approaches using machine learning model (DT), evaluating the IDS framework and point out the classic ML algorithms for anomaly detection and compared the decision tree results with other ML models. The research [66] presented a hybrid (ML+DL)IDS models that combine multiple modalities and parallel processing for enhanced detection and evaluate on benchmark datasets. [67] conducted a deep learning-based IDS framework built on deep convolution neural network and evaluate the performance using bench marks datasets to test IDS dependability under different conditions and shows good accuracy compared to other ML classifier. [68] conducted a systematic literature review for IDS models, analysis IDS studies, highlight commonly used ML techniques for intrusion detection, precisely discuss about datasets and challenges. [69] conducted a systematic literature review on host-based intrusion detection system, review studies that uses ML, statistical, heuristic approaches to host level intrusion detection, analyzing the applicability and challenges. The study [70] presented a systematic literature review of swarm intelligence (SI)-based IDS approaches and review algorithms Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Artificial Bee Colony(ABC), Genetic Algorithms(GA) applied on intrusion detection system and discus the SI techniques in term of feature selection, rule optimization and anomaly detection.

The most important thing to understand about the accuracy presented across different studies is that they cannot be directly compared without considering underlying dataset characteristics. Each standard dataset, such as NSL-KDD, UNSW-NB15, and CICIDS2017, is significantly different in terms of attack types, feature selection, and class imbalance ratios. Moreover, data cleansing techniques like feature selection and scaling techniques also vary among studies, affecting reported performance metrics. As a result, these comparisons should be interpreted carefully, as they do not provide a standardized evaluation framework. Future research should focus on establishing unified benchmarking protocols and cross-dataset evaluations to ensure transparency and stable-performance comparisons of the IDs approach.

Dataset bias and limitations

The Dataset NSL-KDD was used in our analysis because it is a widely used standard for evaluating ID models. The DNN models often perform well on NSL-KDD due to its good-quality features and the ability of deep architectures to capture complex nonlinear relationships across multiple dimensions. Although it is important to understand that NSL-KDD has inherent limitations, such as class imbalance, lack of representation of modern attacks (e.g., IoT, zero-day), and static traffic features, which may lead to dataset-specific optimization rather than real-world generalization. Therefore, future work

should focus on validating IDS models on more realistic and diverse datasets such as UNSW-NB15 or CICIDS2017 to ensure robustness and practical applicability.

Table 6 shows a list of different ML approaches for intrusion detection along with commonly used security datasets. According to [24, 57], ML models are considered supervised learning, where classifiers learn from labeled datasets before making predictions based on fresh data.

Limitations and challenges of datasets and comparison with current survey

Limitations

(i) Duplicate record causes classifier bias, an unbalanced dataset and a lack of realism.
(ii) Deficient in implementation and growth limitation and an intrinsic problem difficulty and understandability.
(iii) Resource-intensive and training time, an overfitting issue and lack of understandabilityand scalability risk.
(iv) Computational cost and class imbalance
(v) Outdated and limited diversity in attack type, and unable to represent encrypted network traffic and not suitable for real-time.
(vi) Outdated for most recent attacks and limited view to zero-day attacks
(vii) Small dataset size and unable to include encrypted traffic scenarios
(viii) The dataset is outdated, unable to tackle current issues, and has duplicated records and synthetic attacks
(ix) Complex feature space for optimal model performance
(x) Imbalance class and preprocessing requirements to handle mixed data types
(xi) Not used publicly or standard available dataset, and a lack of diversity
(xii) System call datasets limited to the real world and the absence of cloud attack senior
(xiii) Focus on HTTP attacks and static dataset and No encrypted traffic

Challenges

(i) Obsolete threats are unable to represent modern cyber attacks and the overfitting risk
(ii) Imbalanced dataset and large number of attributes and real-time detection and evolving cyber threats
(iii) Resource-intensive, training time and overfitting issue, and lack in understandability and scalability risk
(iv) Lack of explainability and dataset-specific tuning
(v) Model computational cost and imbalanced data and scalability and generalization issue
(vi) Information loss and unable to manage nonlinearity, and poor model generalization and interpretation
(vii) Execution cost and overfitting issue, scalability, and interpretability issue
(viii) Real-time processing issue and volatile nature of cloud and lack of accurate data labeling
(ix) Issue related to scarcity, high false positive rate, and issues faced in feature engineering
(x) Unable to feature engineering and issue to limited attack coverage, and scalability and generalization.

(xi) Scalability, generalization, and resource constraint;

Comparison with current survey

(i) This survey analyzes modern datasets and comparative analysis of modern ML-based techniques
(ii) Dataset veracity and future research gaps and cover systematic review methodology
(iii) Provide empirical datasets comparison and PRISMA-based structured methodology, and focus on conceptual benefits and challenges of ML in cyber security without practical development
(iv) The survey paper explores ML-based methods vs traditional methods for IDS, and focuses on a structured approach, comparing multiple datasets
(v) Focus comparative discussion on multiple features selection techniquesand dimensionality reduction techniques of ML highlight advanced model like deep and ensemble learning
(vi) The paper's scope is broader than the previously done and also covers the latest dataset used in IDS.

Table 7 describes the common ML approaches and gives a brief synopsis of ML approaches, including time complexity, advantages and disadvantages, proposed year, and reference number, and Table 8 shows the comparison with our survey paper. The most popular machine learning methods for NIDS implementation are reviewed in the subsequent subsections.

Support vector machines

An ultimate successful and commonly utilized ML model for cybersecurity-related tasks is known as support vector machines (SVM), particularly for IDS. The limits marked on both sides of the hyperplane are used to categorize and divide data into two classes. Increasing the limit as well as the division among hyperplanes will increase the classification accuracy of a data point. Support vector points are the data points that lie on the edge of the hyperplane. SVMs may be separated into 2 main groups: linear or non-linear, and determined through a Kernel function. Depending on the detection type, it could be one class or multiple classes [99, 100]. For optimal learning of dynamic behavior, SVM requires training at various time intervals and demands a significant amount of memory for processing and training. The classifier's performance additionally relies on parameters and the kernel function. The benefits of SVM with respect to IDS are as follows.

- Effective in high-dimensional spaces: It can detect complex intrusion patterns where various network traffic features are included (e.g., packet size, duration, protocol type, etc.).
- Strong generalization ability: Through maximizing the margin among classes, SVM can handle new/unknown attack types that may not be used in the training set. It prevents overfitting.
- Kernel trick for non-linearity: To tackle non-linear decision boundaries, it used the kernel trick that enables SVM to classify complex, non-linear patterns in network traffic because attack signatures and normal network behavior may not be linearly separable

Table 7 An overview of the commonly used ML techniques and their complexities in the context of intrusion detection

Ref. #	Year	Model	Time complexity	Description	Limitation
[85]	2021	SVM	$O(n^2)$	i. It is a good classifier based on statistical techniques and uses a multidimensional hyperplane to classify samples and define a boundary between points in the plane ii. It is commonly used in classification and regression	i. It is unable to manage large datasets effectively ii. It is not cost-effective
[86]	2023	DT	$O(mn^2)$	i. It is a popular and efficient technique for classification ii. It takes less computational time	i. It is challenging to alter the data without changing the structure, time-consuming, and complex.
[87]	2019	ANN	O (emnk)	i. It is adaptive and composed of interconnected artificial neurons ii. The next input layer completely depends on the previous layer's output	i. The black box model defines no connection between input and output variables ii. It consumes huge costs and is time-consuming
[88]	2023	Random Forest	O (Mmlogn)	i. It is made up of many decision trees ii. The model's ultimate result will be the forecast with the most votes	i. Computational cost is higher ii. Very slow prediction generator
[89]	2023	Naive Bayes	O (mn)	i. A probabilistic classifier that takes less execution time ii. It assumes that a feature is completely independent of all other features	i. To acquire decent results, a lot of data is needed ii. If a category from the test data set is absent from the training data set, it is assigned 0.
[90]	2017	K-means	O (kmni)	i. It initially starts by taking random centroids and refines the centroids iteratively until the final cluster analysis	i. It completely depends on starting centroids ii. It is an inefficient way of clustering for different cluster sizes
[91]	2024	RNN	–	i. It is an effective model for sequential data and rapidly absorbs sequential events ii. It is available in different forms like LSTM	i. This is complicated for training the network ii. While modeling long sequences of data, faces short memory issues
[92]	2024	DBN	$O\ (m\sum I^L(I1J^1))$	i. It achieves higher performance because of the addition of layers ii. It has the ability to handle noisy data iii. Hidden layers are effectively used	i. It consumes more hardware resources ii. It is unable to provide an explanation for decisions iii. Due to the higher time consumption of additional layers

- Versatility: SVM is used for various types of intrusion detection tasks, such as identifying specific types of attacks (DoS, DDoS, probe attacks, etc.) due to its versatile nature, either binary classification (attack vs. no attack) or multi-class classification (classifying different types of attacks).

Despite SMV's better results for intrusion detection, it has the following drawbacks.

- Computational complexity and scalability: IDS often deals with huge volumes of network traffic, which makes SVM computationally expensive. Therefore, actual intrusion identification in huge-scale networks might not be possible for SVM.

Table 8 Comparison table

Ref.	Comparison with our study
[93]	Conducted research based on a generic ML pipeline and focused on IoTs, cloud-based IDS, but the current survey paper covers ML-based IDS on computer networks and mobile devices, ML techniques, dataset, evaluation metrics, time complexity, and challenges
[94]	conducted a study narrowly focused on IoT and cloud-based IDs, and a lightweight ML algorithm, covering supervised, unsupervised, and hybrid approaches. This study uses PRISMA-based methodology, details dataset comparison with limitations and challenges, traditional IDS vs modern ML approaches
[95]	Conducted research work based on the collection of multiple original research articles, and covers a much broader range of topics within the IoT era, while this paper covers systematic literature developed to provide a detailed synthesis and analysis of previous research in a specific domain.
[96]	Conducted study and the work is a collection of individual research studies and contains a very wide scope in nature, while this work is a single, systematic literature review, it covers a comprehensive overview and synthesis of previous research, trends, challenges, and future direction within a specific domain.
[97]	conducted a study on an ensemble IDS model (AdaBoost + RF + LR) with high training complexity and better results, while the current review provides a systematic PRISMA-based survey that analyzes ML and DL techniques across multiple domains, compares datasets with limitations, evaluates time complexity, and mentions future research directions
[98]	Conducted a research study based on general intrusion detection system and intrusion prevention system with classical ML models and traditional datasets, but this survey focuses on delivering a systematic PRISMA-based survey that spans ML and DL techniques across multiple domains, includes dataset and time-complexity analysis, and mentions gaps and future-oriented directions

- Sensitivity to parameter tuning: It is highly dependent on proper tuning, which is time-consuming and challenging for dynamic environments like network security.
- Struggles with imbalanced data: It struggles while dealing with highly imbalanced data, which is common in the case of IDS.

Decision tree

An ML method that depends upon the recursive tree structure is called a decision tree (DT). Three components make up DT: a path, an intermediate or root node, and a leaf node. The object is denoted through the tree's root or intermediary node. Every divergence path inside the tree displays the potential values of the parent node. The predictive category characteristic is correlated with the leaf node. If-then rules serve as an additional representation of the resulting tree. The best potential intermediate node has been further selected during the tree's creation through information gain and entropy measures. ID3 and C4.5 are regarded as crucial decision tree algorithms [101]. In order to identify network data flows as either legitimate or malicious, DT is utilized for intrusion detection. It can manage massive volumes of data and facilitate real-time intrusion detection [102]. The DT model carries the following advantages for IDS.

- Handles both categorical (protocol types) and numerical (packet sizes) data
- Provides fast classification
- Handle complex patterns and non-linear relationships among input features and the target class, which is crucial in detecting sophisticated and stealthy attacks.

The DT model is limited by the following factors.

- Prone to over-fitting and complex tree
- Bias toward dominant classes
- Harder to detect complex attacks

- Lack of incremental learning: It requires retraining from scratch when a new attack is found.

K-nearest neighbor

It is a supervised method that does not require a trained model; instead, it employs lazy learning. It retains the training data instances in n-dimensional space instead. The most prevalent label is returned as a forecast after analyzing the K points that are near the testing data points for categorization. Neighbors are denoted by K, which is examined in order to categorize new points in data. Fast KNN is the improved version that is used in NIDS to improve performance and expedite the classification process [103, 104]. The pros of KNN are as follows.

- Handles high-dimensional data (multiple features, e.g., packet size, source IP, protocol type)
- Flexible for new attacks (due to non-parametric and instance-based nature)
- Handles multiclass and multi-label attacks
- Performs well in noisy and incomplete data scenarios

The KNN model has the following limitations for IDS.

- High computational cost
- Selection of K (number of neighbors) is crucial as it may either miss attacks (underfitting) or raise too many false alarms (overfitting).
- Limited interpretability for large datasets

Random forest

It is a kind of ensemble learning, which generates a problem hypothesis by combining several classifiers to establish a typical outcome. It is used for classification and is also referred to as a random decision forest [86]. In order to create an RF model that produces reliable predictions, many decision trees are typically combined. In terms of classification, it is more accurate [86, 88, 105]. Although its classification accuracy is higher, its real-time prediction speed and complexity are slower. Using individual tree output as either normal or attack, RF is utilized for predicting the classes of data in order to detect network intrusions. The class with the highest votes has been selected to provide the final forecast. The benefits of RF concerning IDS are given below.

- High accuracy due to robust ensemble mechanism, RF accurately detecting intrusions, even in complex environments.
- Resilience to overfitting that suited high-dimensional IDS datasets.
- Handle imbalanced data (provides high detection rates for rare or subtle attacks)

The drawbacks of RF concerning IDS are given below.

- Complexity in hyperparameter tuning
- Sensitivity to noisy data
- High computational cost

Naive bayes

Thomas Bayes proposed Bayes' theorem, which assumes the qualities are independent, and is the foundation of the probabilistic classification technique known as Naive Bayes (NB). An NIDS that depends on the NB model has been presented in Panda et al. [89, 106]. This categorization model measures the likelihood that the hypothesis is correct. After assuming that the provided dataset relates to the specific class. In order to handle highly linked items and massive volumes of network data, intrusion detection additionally uses a different form of NB called Hidden Naive Bayes (HNB). The Naive Bayes algorithm requires little data for training, is easy to use, and is scalable. The following are NB's benefits in regard to IDS.

- Perform Well with small data scenarios (when gathering a vast amount of labeled intrusion data is difficult)
- Handle missing data
- Performs well for text data tasks like spam detection

The following are NB's drawbacks in regard to IDS.

- Assumption of feature independence
- Sensitivity to class imbalance
- Handling continuous data

Extreme gradient boost

For classification and regression, machine learning techniques like extreme gradient boost (XGBoost) have been employed. It is a gradient-boosting technique that has been implemented in extraordinarily effective ways and in an adaptable manner. Currently, it is employed to create NIDS models for classification that are incredibly quick and precise. An XGBoost-based classification approach for NIDS has been established by Jiang et al. that determines the ideal structure of XGBoost by particle swarm optimization (PSO) [107]. The pros of XGBoost concerning IDS are as follows.

- Provides high accuracy
- Efficient handling of imbalanced data
- Handle large IDS datasets efficiently due to parallel processing.

The cons of XGBoost concerning IDS are as follows.

- Computationally intensive (tuning hyperparameters)
- Overfitting risk

Artificial neural network

The combination of forward pass and back-propagation loops has been utilized to train artificial neural network (ANN) models. In feed-forward, each node of a buried layer receives data input. Every node in the output and hidden layers has its activation value monitored. A classifier's performance is impacted by the activation function. The difference between the predicted value and the network output is used to compute the error. Using the guardian descent technique, this difference has been transmitted back to the input layer in back-propagation that controls weights among the output and hidden nodes. This process has been repeated until the target threshold is achieved [87, 108].

An ANN holds the following advantages for IDS.

- It can identify new attack patterns and abnormalities because of its capability of learning through data and adapting without the requirement for predefined rules.
- After training, the ANN can identify new, unseen attacks, making them effective against zero-day attacks.
- Real-time intrusion detection

Despite its better results, ANN has the following limitations.

- Training data dependency (requires large, well-labeled datasets)
- Black-box nature (difficult to interpret how decisions are made)

Deep belief networks

An efficient DNN model that employs an unsupervised greedy method is known as a deep belief network (DBN). In order to help the human brain comprehend complex information and recognize intricate patterns, DBN has been developed. DBN is also referred to as a stack of obligatorily generative Restricted Boltzmann machines. Additionally, nodes on the same network layer cannot communicate with one another in DBN. Every node in the DBN is attached to all nodes from the last and next layers [109–111].

Table 9 contains an overview of the state-of-the-art proposed ML approaches for intrusion detection, along with datasets, sub-domains, attack types, and year of publication.

Evaluation metrics for IDS

For calculating the efficiency of ML-based IDS approaches, numerous metrics and parameters have been utilized. Every learning task has its own importance and its various performance measurement metrics. One formal method of presenting the learning model's information is using an error matrix or confusion matrix. It is the tabular representation of a scheme's prediction or classification performance. The outcome of a binary classification into four distinct groups is displayed in the confusion matrix. True-negative-rate (TNR), True-positive-rate (TPR), false-negative-rate (FNR), and false-positive-rate (FPR) values are the output of a classifier that are used to further build additional metrics. For that reason, it is impossible to pinpoint which parameters are more crucial for a certain intrusion detection class. Cybersecurity categorization methods are frequently evaluated using the following criteria or terms:

True positive rate

The percentage of accurately anticipated assaults relative to all attacks is what it refers to. In the event that all intrusions have been detected, then $TPR = 1$, is remarkably unmatched in the case of IDS. The rate of detection and sensitivity are others name for TPR. TPR is expressed as in Eq. 1.

$$\mathbf{TPR} = \frac{\mathbf{T}.(+\mathbf{ve})}{\mathbf{T}.(+\mathbf{ve}) + \mathbf{F}.(-\mathbf{ve})} \tag{1}$$

Table 9 Overview of state-of-the-art proposed ML Techniques for intrusion detection with datasets

Ref.	Title	Year	Sub-domain	Model	Dataset	Attack types
[112]	Machine-Learning-Based Feature Selection Techniques for Large-Scale Network Intrusion Detection	2014	Feature-selection	RF-FSR, RF-BER	KDD-99	DoS, U2R, Probing
[113]	Machine-Learning-Based Feature Selection Techniques for Large-Scale Network Intrusion Detection	2014	–	Garrett's ranking	NSL-KDD	DoS, U2R, Probing
[114]	Implementation of Network Intrusion Detection System using Variant of Decision Tree Algorithm	2015	Misuse-based	NB	NSL-KDD	–
[115]	Classification of Attacks Using Support Vector Machine (SVM) on KDDCUP'99 IDS Database	2015	–	RBF-SVM	KDDCup99	–
[76]	Intrusion Detection System using PCA and Fuzzy PCA Techniques	2016	–	PCA, K-NN	KDD 99	R2L, Probing
[116]	Enhancing performance of anomaly-based intrusion detection systems through dimensionality reduction using principal component analysis	2016	Anomaly-based	MLP, PCA, NB, SVM	NSL-KDD	DoS, U2R, R2L
[106]	Analysis of Machine Learning Techniques Based Intrusion Detection Systems	2016	Anomaly-based	SVM	NSL-KDD	–
[117]	An effective combining classifier approach using tree algorithms for network intrusion detection	2017	Hybrid-based	Decision Tree	KDD	–
[118]	Incremental k-NN SVM Method in Intrusion Detection	2017	–	SVM, K-Mean, K-NN	KDD99	DoS, U2R, Prob, R2L
[119]	An Effective Intrusion Detection Classifier Using Long Short-Term Memory with Gradient Descent Optimization	2017	–	LSTM	KDDCup99	–
[120]	Network Intrusion Detection Using Kernel-based Fuzzy-rough Feature Selection	2018	–	NB, Ada-Boost, RF	KDD99	DoS, Probing, R2L
[121]	Identification of malicious activities in industrial IoT based on deep learning models	2018	Anomaly-based	ANN	UNSW-15	DoS, Probing, R2L
[122]	A hybrid technique using binary particle swarm optimization and decision tree pruning for network intrusion detection	2018	Hybrid-based	Decision Tree	KDDCup99	–
[123]	A New Swarm Optimization Based Technique for Anomaly Detection	2019	Anomaly-based	ANN	NSL-KDD	–

Table 9 (continued)

Ref.	Title	Year	Sub-domain	Model	Dataset	Attack types
[110]	Intrusion Detection for IoT Based on Improved Genetic Algorithm and Deep Belief Network	2019	Anomaly-based	DBN, RF	NSL-KDD	–
[124]	Intrusion Detection for IoT Based on Improved Genetic Algorithm and Deep Belief Network	2019	Anomaly-based	RF	NSL-KDD	–
[125]	Deep Learning Approach for Intelligent Intrusion Detection System	2020	–	DNN	NSL-KDD	DoS, Probing, R2L
[126]	SS symmetry Learning and Knowledge Model	2020	–	NB, DT, RF	KDD-99	DoS, Probing, R2L
[107]	Leveraging Machine Learning Approach to Setup Software-Defined Network (SDN) Controller Rules During DDoS Attack	2020	–	J48, SVM, NB	Customized	DDoS
[127]	Analysis of intrusion detection in cyber attacks using DEEP learning neural networks	2021	–	CNN, SOM	transformed	U2R, DoS
[85]	Machine learning methods for cyber-security intrusion detection: Datasets and comparative study	2021	Max-Min	SVM	UNSW-NB15	–
[128]	MapReduce-based intelligent model for intrusion detection using machine learning technique	2022	MR-IMID	ANN	Large dataset	DoS
[129]	Network intrusion detection system using Machine learning approach	2022	KNIME	SVM, RProp, DT	CICIDS2017	Benign, DoS, Bot, Web attack
[130]	Ensuring network security with a robust intrusion detection system using ensemble-based ML	2023	Ensemble-based	RF	Public datasets	–
[131]	Cloud-Based Intrusion Detection Approach Using Machine Learning Techniques	2023	Anomaly-based	RF	NSLKDD, Bot-IoT	–
[40]	An efficient feature selection assisted network intrusion detection system using machine learning	2024	Singular-value decomposition	M-Multi SVM	CSECIC-IDS2018, UNSW-NB15	Brute-force, heart bleed, Botnet, shellcode
[132]	Stochastic Gradient Descent Intrusions Detection for Wireless Sensor Network Attack Detection System Using Machine Learning	2024	Decision-based	GNB, SGD	WSN_DS	WSN-intruder

False positive rate

The number of usual events that are mistakenly categorized as an attack on the entire usual occurrences is presented by FPR. The FPR is a measure of the percentage of valid activities that the system misclassifies as incursions. F. (+ve)s, or genuine acts that are labeled as malicious, are divided by the sum of F. (+ve)s and T. (-ve)s, or legitimate actions that are correctly detected as non-intrusive, to determine this metric. A false

alarm rate (FPR) with a high value means that IDS produces a lot of false alarms, which might cause security staff to get alert fatigued and raise the possibility that real threats will go unnoticed. Therefore, reducing the FPR is crucial to improving an IDS's efficacy and usability. FPR is expressed in Eq. 2.

$$\mathbf{FPR} = \frac{\mathbf{F}.(+\mathbf{ve})}{\mathbf{F}.(+\mathbf{ve}) + \mathbf{T}.(-\mathbf{ve})} \tag{2}$$

False negative rate

The number of abnormal objects that are mistakenly classified as normal by the IDS is denoted by FNR. It may be computed as the fraction of the entire abnormal events (sum of F. (-ve)s and T. (+ve)s) to the number of missing abnormal items (F. (-ve)s) as shown in Eq. 3. Systems may become vulnerable to assaults if an IDS has a high FNR, which means it misses a large percentage of real threats. Since a low FNR indicates that the system correctly detects the majority of intrusions, improving the overall security posture, this rate is essential for assessing an IDS's efficacy.

$$\mathbf{FNR} = \frac{\mathbf{F}.(-\mathbf{ve})}{\mathbf{F}.(-\mathbf{ve}) + \mathbf{T}.(+\mathbf{ve})} \tag{3}$$

True negative rate

The fraction of accurately categorized –ve samples (i.e., legal activities accurately identified as non-intrusive) to the entire genuine –ve events in the data is called specificity, or TNR. A higher TNR depicts the algorithm performing efficiently and that the IDS is successfully differentiating between genuine activity and real intrusions. The amount of T. (-ve)s divided by the total of F. (+ve)s and T. (-ve)s yields TNR as depicted in Eq. 4. A high TNR is preferable as it indicates that the system reduces false alarms, freeing up security staff to concentrate on real threats while preserving operational effectiveness.

$$\mathrm{TNR} = \frac{\mathbf{T}.(-\mathbf{ve})}{\mathbf{T}.(-\mathbf{ve}) + \mathbf{F}.(+\mathbf{ve})} \tag{4}$$

Accuracy

A fraction of accurately identified events among all events is known as accuracy, and it serves as a gauge for how well the IDS detects typical or unusual traffic patterns [133]. Accuracy is a crucial performance metric that shows how well the system can identify malicious or benign network data. It may computed by dividing the entire number of events (T. (+ve)s, F. (+ve)s, T. (-ve)s, and F. (-ve)s) by the sum of T. (+ve)s (attacks that were successfully recognized) and T. (-ve)s (regular traffic that was correctly identified) as shown in Eq. 5. By reducing both undiscovered assaults and false alarms, an increased accuracy indicates that the IDS successfully distinguishes between invasive and authorized activity, improving the overall security posture. To offer a thorough evaluation of the IDS's performance, accuracy should be taken into account in conjunction with other ways, like recall, precision, and the FPR, even if accuracy is crucial.

$$\mathrm{Accuracy} = \frac{\mathbf{T}.(-\mathbf{ve}) + \mathbf{T}.(+\mathbf{ve})}{\mathbf{T}.(+\mathbf{ve}) + \mathbf{T}.(-\mathbf{ve}) + \mathbf{F}.(+\mathbf{ve}) + \mathbf{F}.(-\mathbf{ve})} \tag{5}$$

Precision

The proportion of anticipated assaults to the entire sample that have been forecast is known as precision. In particular, it divides the number of true positive detections (actual attacks accurately recognized) by the sum of T. (+ve)s and F. (+ve)s (total number of predictions made as attacks) to determine how accurate the IDS is in spotting harmful activity. High accuracy means that the majority of the actions that are identified are malicious, indicating that the IDS is effective in reducing false alarms. This is essential to preserving system credibility and allowing security staff to concentrate on real dangers without being distracted by pointless alarms. Mathematically, it can be defined as follows.

$$\text{Precision} = \frac{\mathbf{T}.(+\mathbf{ve})}{\mathbf{F}.(+\mathbf{ve}) + \mathbf{T}.(+\mathbf{ve})} \tag{6}$$

The precision of various models is depicted in Fig. 7a while the recall of some models is given in Fig. 7b. Figure 7c provides the accuracy of different proposed ML-based IDS models.

F measure

Recall and precision-based harmonic mean is denoted by F.1-score/F-measure. It is a statistical method for assessing a system's correctness while accounting for both precision and recall. It offers a fair statistic that aids in evaluating the effectiveness of the system by taking seriously the trade-off between F. (+ve)s and F. (-ve)s for IDS. Recall assesses how well the system can identify real intrusions among all instances of genuine intrusions, whereas precision gauges how well the IDS detects harmful activity among all marked cases. The F-measure is especially helpful in situations where both false positives—incorrectly flagging legitimate actions—and false negatives—missed

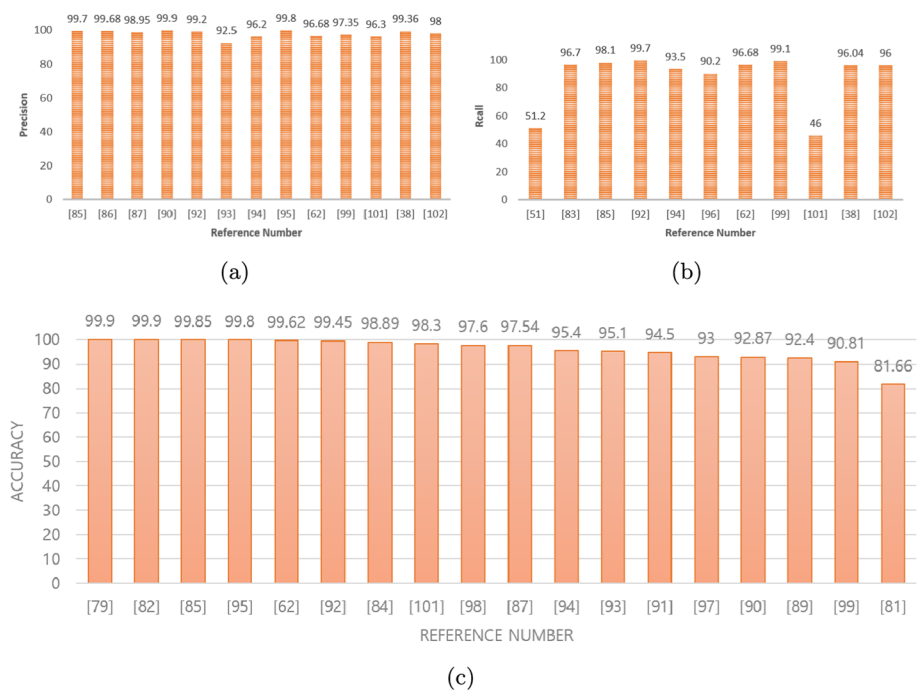

Fig. 7 Results of various ML models for IDS from existing works, (**a**) Precision, (**b**) Recall, and (**c**) Accuracy

intrusions—can have serious repercussions. It enables security experts to assess how well an IDS is keeping an environment safe while lowering the possibility of missing possible threats [134]. It is represented in Eq. 7.

$$F - Score = 2\left(\frac{Prec * Rec}{Prec + Rec}\right) \tag{7}$$

Error rate

The proportion of observations in the datasets that have been misidentified among all samples is denoted by the error rate. A lower error rate indicates higher classifier performance and is preferred. It is represented in Eq. 8.

$$E - Rate = \frac{F.(+ve) + F.(-ve)}{T.(-ve) + F.(-ve) + T.(+ve) + F.(+ve)} \tag{8}$$

G mean

G-mean is the measurement approach that the classifier uses to evaluate the actual predicted values. Accuracy doesn't display the right image for +ve observations when there are more –ve items than $+ve$ items. Equation 9 shows the mathematical representation of G-mean.

$$G - Maen = \sqrt{\frac{TP}{(TP + FN)\,(\frac{XTN}{TN} + FP)}} \tag{9}$$

Receiver operating characteristics curve

The receiver operating characteristic ROC) curve is the most often used graphical representation that plots the values of TPR (y-plane) versus FPR (x-plane) to provide an overview of all threshold performance.

Area-under-curve

The region that falls under ROC is known as the area under the curve (AUC), and its value lies between 1/2 to 1. An ML classifier having a good AUC value performs more efficiently.

Mean squared error

The mean of the error square or difference between the classifier's forecast and real values is known as mean squared error (MSE). Fewer values of MSE depict higher classifier performance and are preferred.

Mean absolute error

The mean absolute error (MAE) may be calculated simply by averaging the absolute error or difference between the classifier's predicted and real values. A lower MAE value indicates powerful classifier performance and is preferred.

Root mean square error

The root mean square error (RMSE) may be obtained by computing the root-mean-square value of MSE. Better classifier performance is indicated by a lower RMSE score.

Table 10 depicts the pros and cons of different evaluation metrics that are commonly used for IDS.

Table 11 depicts the performance evaluation values for numerous SOTA Intrusion Detection Techniques with datasets and attack types.

Challenges and research gaps

The practical implementation of ML-based intrusion detection systems faces many crucial challenges, which are grouped into three ways. (i) Dataset-related challenges: Standard datasets such as NSL-KDD and UNSW-NB15 faces issues like class imbalance, outdated attack types, and lack of real-time traffic diversity, which hide abstraction and bias detection accuracy [135], for example, NSL-KDD is a conventional attack failed to trained models and has been the reason to perform poorly on modern threats such as botnets and advanced persistent threats in CICIDS2017 [136]. (ii) Algorithmic challenges: Normal and common issues like high false positive rates, overfitting, and the inability of models to adapt to zero-day attacks. While Deep learning models are powerful and data hungry, sensitive to adversarial samples, making them unrealistic for dynamic attack environments [137–140]. (iii) Deployment-related challenges: Real-world intrusion detection models need to be efficient in terms of computation and scalability. Although costly in the context of resource models such as deep neural networks, which demand high GPU processing power, which is impractical for edge devices and IoT architecture [141]. There are lots of performance gaps observed in already published papers, such as reduced detection accuracy of SVM and decision trees on UNSW-NB15 and NSL-KDD, which are directly connected to these challenges. Addressing these limitations requires research into adaptive IDS frameworks, federated learning for

Table 10 Pros and cons of evaluation metrics for IDS

Metric	Pros	Cons
TPR	Evaluates sensitivity	Can be misleading if the negative class is more important
TNR	Calculate the capability to precisely detect negative instances	May hide false negatives (rare attack)
FPR	measure the cost of false alarms (highly sensitive systems)	Struggles with imbalance class
FNR	Provides highlights of missed positive instances	Less useful in highly imbalanced datasets where the positive class is rare
Accuracy	Depicts an overall view of model performance	Can be deceptive in imbalanced datasets (have a dominant class)
Precision	Useful where false alarms are costly.	Doesn't account for false negatives
F-Measure	Helpful while both FP and FN are important.	Ignores true negatives
G-Mean	Ideal for imbalanced datasets.	Fails to provide information on the differences between FPR and FNR.
ROC	Visualizes the trade-off between TPR and FPR.	Provide misleading in imbalanced datasets
AUC	Summarizes the performance of the classifier across all threshold settings	less informative when the problem involves imbalanced classes
MSE	Evaluate prediction errors in regression tasks.	Not ideal for classification problems
MAE	Provides calculation of mean size of prediction errors	Not penalize large errors strongly
RMSE	Similar to MSE, with more interpretable	Less suited to classification problems

Table 11 Performance efficiency of SOTA IDS techniques with datasets and attack types

Ref.	Dataset	Attack type	Accuracy (%)	Precision (%)	Recall (%)
[112]	KDD-99	DoS, U2R	99.90	–	–
[113]	NSL-KDD	DoS, U2R	96.4, 96.14	–	–
[114]	NSL-KDD	–	81.66	–	–
[115]	KDDCup99	–	99.9	–	–
[76]	KDD 99	R2L, Probing	–	97.8, 93	51.2, 50
[116]	NSL-KDD	DoS, U2R	97.1, 97.3	–	96.7, 97.9
[106]	NSL-KDD	–	98.89	–	–
[117]	KDD	–	99.85	99.7	98.1
[118]	KDD99	DoS, U2R	–	99.68	–
[119]	KDDCup99	–	97.54	98.95	–
[120]	KDD99	DoS, U2R	91.03, 99.89	–	–
[121]	UNSW-15	DoS, U2R	92.4	–	–
[122]	KDDCup99	–	92.87	99.9	–
[123]	NSL-KDD	–	94.5	–	–
[110]	NSL-KDD	–	99.45	99.2	99.7
[124]	NSL-KDD	–	95.1	92.5	–
[125]	NSL-KDD	DoS, U2R	95.4	96.2	93.5
[126]	KDD-99	DoS, U2R	99.8	99.8	–
[107]	Customized	DDoS	90.3, 85.3	–	90.2
[127]	Transformed	U2R, DoS	93	–	–
[85]	UNSW-NB15	–	99.62	96.68	96.68
[128]	Large Dataset	DoS	97.6	–	–
[129]	CICIDS2017	Benign, DoS, Bot	90.81	97.35	99.10
[131]	NSLKDD	Cloud-based IDS	98.3	96.3	46.0
[40]	CSECIC-IDS2018, UNSW-NB15	Brute-force, Heart bleed, Botnet, Shellcode	97.80, 85.88	99.36, 89.36	96.04, 85.93
[132]	WSN-DS	WSN-intruder	0.82, 0.95	0.81, 0.98	0.82, 0.96

distributed detection, and standardized benchmarking protocols for transparent evaluations across datasets. ML algorithms are often utilized for making decisions related to illness diagnosis in the medical field or terrorist identification. In such situations, to prevent disastrous outcomes, forecasting cannot be relied upon in a blind manner [104, 142]. While using ML approaches in real-time, crucial applications (e.g., cyber-security, surgical robotics, self-driving cars), it has been challenging to ensure that these systems offer high-level assurances of correctness rather than speed and accuracy [143–145]. For cyberspace security concerns, trustworthy ML models are the most reliable solution. We have investigated the SOTA algorithms and techniques of ML that have been utilized for handling cyber-crimes like IDS. These issues are described in Table 12. In addition, numerous contradictions are addressed, and problems are clarified, providing a solid foundation for talking about other future challenges.

Future work

Although ML-based intrusion detection has gained prominent success, several challenges still remain unaddressed, and provide insight for future research and improvement:

Table 12 Challenges in ML-based IDS

Sr #	Challenges
1	Most research works have been done while using a small number of datasets. There is a major issue: most datasets are out-of-date. To tackle this challenge, new datasets must be compiled.
2	An approach's temporal complexity, along with the real-time detection rate setting, should be taken into account; it is also an addressable issue.
3	It is important to consider advanced computation costs and detection speed.
4	Studies either utilized a single classifier or a limited number of classifiers. It's unable to properly state which ML technique performs at its best. There is another severe challenge: testing several ML models because, by including hostile text sequences, an attacker can alter the message.
5	ML approaches require large amounts of high-performance data and resources to train the models.
6	Moreover, data collected from diverse sources (structured, unstructured, semi-structured) can lead to security issues.
7	Instead of concentrating just on efficiency and precision, it is critical to ensure that models provide top-level accuracy guarantees.
8	Deep learning models may struggle to easily model several complex data modalities.
9	Distributed and Parallel DL models are usually not employed in intrusion detection.
10	Maintaining an object's dependability in the face of potential errors can seem challenging.
11	With the passage of time, progress, and innovation in cybersecurity, the progression of attacks is growing rapidly.
12	Confidentiality and protection of data are crucial because data are gathered from diverse sources. This raises the issue of whether big data should be used for security or for something else.

Real-world and diverse dataset

Previous benchmark datasets, such as KDD'99 and NSL-KDD, are outdated and insufficient to represent network traffic and emerging attacks like IoT-based threats. Future work should focus on developing detailed and heterogeneous datasets that include real-world scenarios from cloud computing and IoT networks to enhance the model generalizability.

Real-time and scalable IDS

Current ML-based IDSs are unable to meet the requirements of high-speed networks and large-scale distributed systems. So, there is a need to prioritize the design of low-latency detection techniques and an edge IDS framework to make real-time intrusion detection without losing correctness.

Integrated contextual detection

Combining signature-based, anomaly-based, and ML-based approaches can improve detection exactness and lessen false positives. In future IDS solutions, context-aware and behavior-driven identification techniques which is helpful for users and devices for accurate attack detection.

Privacy-preserving IDs

Data privacy is an ongoing challenge that needs to be fixed in the future to employ privacy-preserving ML techniques such as federated learning to permit collective training across multiple organizations without sharing original data.

Human activity recognition

ML-based approaches are expected to enhance the research on recognizing human activity. This involves recognizing and predicting human activities using data from smart devices, which can be beneficial for security and other applications.

Exploring unique approaches

To further enhance the domain, future work must focus on investigating unique techniques for anomaly and intrusion detection. This includes developing novel algorithms and methodologies that can address the specific challenges associated with IoT and big data security.

Incorporating with emerging technologies

To fully utilize the blockchain for data integrity, AI-based automation for adaptive attack reply and edge computing for decentralized network security can be helpful and more resilient in dynamic environments like smart cities, industrial control systems, and 5th generation.

Therefore, addressing these future challenges and research directions will be crucial for the continued advancement of ML techniques in cybersecurity. By focusing on these areas, researchers can develop more efficient and successful systems for unusual behavior and intrusion detection, particularly for IoT and the security of big data.

Conclusion

Cybersecurity still remains a crucial and global challenge, as conventional security measures such as firewalls and anti-virus programs are unable to detect advanced and existing unknown threats. This survey paper highlights the importance of ML-based IDS in addressing these limitations. We present a comprehensive review of ML-based techniques employed in IDS, including their categories, architecture, commonly used datasets, evaluation metrics, and the time complexity of popular algorithms, lack of real-time adaptability, threat to adversarial attacks, and model interpretability. Although ML-based IDS techniques have presented effectiveness in detecting anomalies and zero-day attacks, challenges such as a high false positive rate and facing issues continuously modifying models still remain unsolved. So, there is no single ML approach that can provide a universal solution and offer hybrid models to provide a more precise security framework.

This study contributes by summarizing state-of-the-art IDS research, identifying key limitations, and outlining the comparative strengths and weaknesses of multiple models. The resultant findings emphasize on the (i) identification of critical gaps, such as lack of standard benchmarking include resilience robustness and clarity in intrusion detection systems, and (ii) a structured research framework and methods across dataset evaluation, lightweight IDS for IoT and more advanced and intelligent ML (iii) practical guide-lines for experts to select suitable algorithms and datasets-based implementation requirements. This ensures that the review not only synthesizes existing work but also gives a clear direction for future research and scalable, privacy-preserving IDS solutions to secure future generations' cyber threats in complex network environments.

Author contributions

HMRuR conceptualization, formal analysis, writing - the original draft. SL data curation, conceptualization, writing - the original draft. MJG methodology, formal analysis, data curation. MAJ software, methodology, project administration. DG funding acquisition, investigation, visualization. MMV visualization, software, investigation. IA supervision, validation, writing - review and editing. All authors reviewed the manuscript and approved it.

Funding

This study was funded by the European University of Atlantic.

Data availability

Not applicable.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]School of Computer Science and Engineering, Yeungnam University, 280-Daehak-ro, Gyeongsan 38541, Republic of Korea
[2]Department of Data Science, The Islamia University of Bahawalpur, Baghdad road, Bahawalpur 63100, Pakistan
[3]Universidad Europea del Atlántico, Isabel Torres 21, Santander 39011, Spain
[4]Universidad Internacional Iberoamericana, Campeche 24560, Mexico
[5]Universidade Internacional do Cuanza, Cuito, Bie, Angola
[6]Universidad Internacional Iberoamericana Arecibo, Puerto Rico 00613, USA
[7]Universidad de La Romana, La Romana, Dominican Republic
[8]Department of Information and Communication Engineering, Yeungnam University, 280-Daehak-ro, Gyeongsan 38541, Republic of Korea

Received: 26 November 2024 / Accepted: 21 October 2025
Published online: 28 November 2025

References

1. Sanou B. Facts and 2017 figures. Available online: https://www.url.com (Accessed: Date) 2017.
2. Union IT, Sector D. Facts and Figures: focus on Least Developed Countries. Available online: https://www.url.com (Accessed: Date), 2023.
3. Vartouni AM, Kashi SS, Teshnehlab M. An anomaly detection method to detect web attacks using stacked auto-encoder. In: 2018 6th Iran. Jt. Congr. Fuzzy Intell. Syst. CFIS 2018, vol. 2018-January, 2018. p. 131–134. https://doi.org/10.1109/CFIS.2018.8336654.
4. Sun N, Zhang J, Rimba P, Gao S, Zhang LY. Data-driven cybersecurity incident prediction: a survey. IEEE Commun Surv Tutor. 2019;21(2):1744–72. https://doi.org/10.1109/COMST.2018.2885561.
5. McIntosh T, Jang-jaccard J, Watters P, Susnjak T. The Inadequacy of Entropy-Based Ransomware Detection, vol. 1. Springer; 2019. https://doi.org/10.1007/978-3-030-36802-9.
6. Cost of a Data Breach Report 2019.
7. Security I. Cost of a Data Breach Report. Accessed 23 April 2024 2023. https://www.ibm.com/downloads/cas/E3G5JMBP
8. Sarker IH, Kayes ASM, Badsha S, Alqahtani H, Watters P, Ng A. Cybersecurity data science: an overview from machine learning perspective. J Big Data. 2020. https://doi.org/10.1186/s40537-020-00318-5.
9. Ambalavanan V. Cyber threats detection and mitigation using machine learning. https://doi.org/10.4018/978-1-5225-9611-0.ch007.
10. al, PO. The comprehensive national cybersecurity initiative, 2009. p. 1–5.
11. Xu Z, Wu Y, Wang S, Gao J, Qiu T, Wang Z. Deep Learning-based Intrusion Detection Systems: a Survey. J ACM. 2025;1(1):1–35 arXiv:2504.07839v2.
12. Retracted: intrusion detection for industrial control systems 2023;2021:9807486.
13. Rahman MM, Al Shakil S, Mustakim MR. A survey on intrusion detection system in IoT networks. Cyber Security and Applications. 2025;3:100082.
14. Pandeeswari GJ. Analysis of intrusion detection using machine learning techniques. In: 2022 Second International Conference on Advanced Technologies in Intelligent Control and Environment Computation Communication Engineering, 2022. p. 1–5. https://doi.org/10.1109/ICATIECE56365.2022.10047057.
15. Ibitoye O, Shafiq O, Matrawy A. Analyzing Adversarial Attacks Against Deep Learning for Intrusion Detection in IoT Networks 2019. arXiv:1905.05137.
16. Chacko AA, Edwin B. Intrusion detection using machine learning techniques: An exhaustive review. In: 2023 9th International Conference on Advanced Computing, Communication Systems, vol. 1. 2023. p. 1586–1589. https://doi.org/10.1109/ICACCS57279.2023.10112831.
17. Singh A. Survey on machine learning applications in cyber security problems. Int J Comput Appl. 2022;7(6):56–61.
18. Gul MJ, Rabia R, Jararweh Y, Rathore MM, Paul A. Security flaws of operating system against live device attacks: A case study on live linux distribution device. In: 2019 Sixth International Conference on Software Defined Systems (SDS), 2019. p. 154–159. https://doi.org/10.1109/SDS.2019.8768590.
19. Jamshidi S, Nikanjam A, Nafi KW, Khomh F, Rasta R. Application of deep reinforcement learning for intrusion detection in internet of things: a systematic review. Internet of Things. 2025. https://doi.org/10.1016/j.iot.2025.101531.
20. Jiang H, Nagra J, Ahammad P. Sok: applying machine learning in security—a survey. 2012.
21. Wijesinghe LS, Silva LNBD, Abhayaratne GTA, Krithika P, Priyashan SMDR. Combating cyber crime using artificial agent systems. Int J Sci Res Publ. 2016;6(4):265–74.
22. Das R, Morris TH, Ph D. Machine learning and cyber security. In: 2017 International Conference on Computer, Electrical and Communication Engineering, 2017. p. 1–7. https://doi.org/10.1109/ICCECE.2017.8526232.
23. Hodo E, Bellekens X, Hamilton A, Tachtatzis C. Shallow and deep networks intrusion detection system: a taxonomy and survey, 1–43.
24. Xin Y, Kong L, Liu Z, Chen Y, Li Y, Zhu H, et al. Machine learning and deep learning methods for cybersecurity. IEEE Access. 2018;6:35365–81. https://doi.org/10.1109/ACCESS.2018.2836950.
25. Li J. Cyber-security meets artificial intelligence A survey. Frontiers Inf Tech Electro Eng. 2018;19(12):1462–74.

26. Ucci D, Aniello L, Baldoni R. Survey of machine learning techniques for malware analysis. Comput Secur. 2019;81:123–47. https://doi.org/10.1016/j.cose.2018.11.001.
27. Apruzzese G. On the effectiveness of machine and deep learning for cyber security. In: 2018 10th International Conference on Cyber Conflict, 2018. p. 371–390. https://doi.org/10.23919/CYCON.2018.8405026.
28. Veiga AP. No Title. March, 2018.
29. Guan Z. When machine learning meets security issues: A survey. In: 2018 IEEE International Conference on Intelligent Safety Robotics, 2018. p. 158–165.
30. Shalaginov A, Banin S, Dehghantanha A, Franke K, Aug CR. Machine learning aided static malware analysis: a survey and tutorial. Cyber Threat Intell. (1):1–37.
31. Sagar BS. Providing cyber security using artificial intelligence: a survey. In: 2019 3rd International Conference on Computer Methodologies and Communications. 2019. p. 717–720.
32. Kwon D, Kim H, Kim I, Kim KJ, Kim J, Suh SC. A survey of deep learning-based network anomaly detection. Clust Comput. 2019;22(s1):949–61. https://doi.org/10.1007/s10586-017-1117-8.
33. Berman DS, Buczak AL, Chavis JS, Corbett CL. A survey of deep learning methods for cyber security. Information. 2019. https://doi.org/10.3390/info10040122.
34. Martínez J, Carla T, Comesaña I, García PJ. Review: machine learning techniques applied to cyber-security. Int J Mach Learn Cybern. 2019;10(10):2823–36. https://doi.org/10.1007/s13042-018-00906-1.
35. Yıldırım T, Genovese A, Scotti F. A comprehensive survey of databases and deep learning methods for cybersecurity. IEEE Syst J. 2021;15(2):1717–31.
36. Molina-coronado B, Mori U, Mendiburu A, Miguel-alonso J. Survey of network intrusion detection methods from the perspective of the knowledge discovery in databases process. IEEE Trans Netw Serv Manage. 2020;17(4):2451–79. https://doi.org/10.1109/TNSM.2020.3016246.
37. Rajapaksha S, Kalutarage H, Madzudzo G, Cheah M. Ai-based intrusion detection systems for in-vehicle networks: a survey. ACM Comput Surv. 2023;55(11):1–40. https://doi.org/10.1145/3570954.
38. Venkatesan S. Design an intrusion detection system based on feature selection using ml algorithms. Math Stat Eng Appl. 2023;72(1):702–10.
39. Saied M, Guirguis S, Madbouly M. Review of artificial intelligence for enhancing intrusion detection in the internet of things. Eng Appl Artif Intell. 2024;127(PA):107231. https://doi.org/10.1016/j.engappai.2023.107231.
40. Turukmane AV, Devendiran R. Multisvm: an efficient feature selection assisted network intrusion detection system using machine learning. Comput Secur. 2024;137(November 2023):103587. https://doi.org/10.1016/j.cose.2023.103587.
41. Sun Z, An G, Yang Y, Liu Y. Franklin open optimized machine learning enabled intrusion detection system for internet of medical things. Franklin Open. 2024;6(November 2023):100056. https://doi.org/10.1016/j.fraope.2023.100056.
42. Klumpner C, Wijekoon T, Wheeler P. New methods for the active compensation of unbalanced supply voltages for two-stage direct power converters. IEEJ Trans Ind Appl. 2006;126(5):589–98. https://doi.org/10.1541/ieejias.126.589.
43. Jang-jaccard J. A survey of emerging threats in cybersecurity. J Comput Syst Sci. 2014;80(5):973–93. https://doi.org/10.1016/j.jcss.2014.02.005.
44. Alazab M, Venkatraman S, Watters P, Alazab M. Zero-day malware detection based on supervised learning algorithms of api call signatures. 2011.
45. Sarker IH. Deep cybersecurity: a comprehensive overview from neural network and deep learning perspective. SN Comput Sci. 2021;2(3):1–16. https://doi.org/10.1007/s42979-021-00535-6.
46. Imran M, Siddiqui HUR, Raza A, Raza MA, Rustam F, Ashraf I. A performance overview of machine learning-based defense strategies for advanced persistent threats in industrial control systems. Computers & Security. 2023;134:103445.
47. Kyaw PH, Gutierrez J, Ghobakhlou A. A systematic review of deep learning techniques for phishing email detection. Electronics. 2024;13(19):3823.
48. Aslam S, Aslam H, Manzoor A, Chen H, Rasool A. Antiphishstack: Lstm-based stacked generalization model for optimized phishing url detection. Symmetry. 2024;16(2):248.
49. Alamri Z, Alhuzali A, Alsulami B, Alghazzawi D. Enhanced phishing website detection using optimized ensemble stacking models. Int J Adv Comput Sci Appl. 2025;16(8):1.
50. Jabbar H, Al-Janabi S. Ai-driven phishing detection: enhancing cybersecurity with reinforcement learning. J Cybersecur Privacy. 2025;5(2):26.
51. Alauthman M, Aldweesh A, Al-Qerem A, Alkasassbeh M, Alateef S, Almomani A. Synthetic content generation impacts on phishing and impersonation attacks. In: Examining Cybersecurity Risks Produced by Generative AI. IGI Global Scientific Publishing; 2025. p. 189–210.
52. George AS. Riding the AI waves: an analysis of artificial intelligence's evolving role in combating cyber threats. Part Univ Int Innov J. 2024;2(1):39–50.
53. Shiravi A, Shiravi H, Tavallaee M, Ghorbani AA. Toward developing a systematic approach to generate benchmark datasets for intrusion detection. Comput Secur. 2011;31(3):357–74. https://doi.org/10.1016/j.cose.2011.12.012.
54. Buczak AL, Guven E. A survey of data mining and machine learning methods for cyber security intrusion detection. IEEE Commun Surv Tutor. 2016;18(2):1153–76. https://doi.org/10.1109/COMST.2015.2494502.
55. Durole PH, Agarwal M. A comprehensive review of advanced artificial intelligence techniques to enhance intrusion detection systems. In: 2024 IEEE Int. Students' Conf. Electr. Electron. Comput. Sci., 2020. p. 1–7. https://doi.org/10.1109/SCEECS61402.2024.10482305.
56. Citation ES. A review of kdd99 dataset usage in intrusion detection and machine learning between 2010 and 2015. 2016;0–21.
57. Agrawal S, Agrawal J. Survey on anomaly detection using data mining techniques. Procedia Comput Sci. 2015;60:708–13. https://doi.org/10.1016/j.procs.2015.08.220.
58. Ring M, Wunderlich S, Scheuring D, Landes D, Hotho A. A survey of network-based intrusion detection data sets. Comput Secur. 2019;86:147–67.
59. Hnamte V, Hussain J. An extensive survey on intrusion detection systems: Datasets and challenges for modern scenario. In: 2021 3rd International Conference on Electrical, Control and Instrumentation Engineering (ICECIE), 2021. p. 1–10. IEEE.
60. Srivastava D, Singh R, Chakraborty C, Maakar SK, Makkar A, Sinwar D. A framework for detection of cyber attacks by the classification of intrusion detection datasets. Microprocess Microsyst. 2024;105:104964.

61. Jayalaxmi P, Saha R, Kumar G, Conti M, Kim T-H. Machine and deep learning solutions for intrusion detection and prevention in iots: a survey. IEEE Access. 2022;10:121173–92.
62. Madhu B, Chari MVG, Vankdothu R, Silivery AK, Aerranagula V. Intrusion detection models for iot networks via deep learning approaches. Meas Sensors. 2023;25:100641.
63. Umer MA, Junejo KN, Jilani MT, Mathur AP. Machine learning for intrusion detection in industrial control systems: applications, challenges, and recommendations. Int J Crit Infrastruct Protect. 2022;38:100516.
64. Dina AS, Manivannan D. Intrusion detection based on machine learning techniques in computer networks. Internet of Things. 2021;16:100462.
65. Azam Z, Islam MM, Huda MN. Comparative analysis of intrusion detection systems and machine learning-based model analysis through decision tree. IEEE Access. 2023;11:80348–91.
66. Shi S, Han D, Cui M. A multimodal hybrid parallel network intrusion detection model. Connect Sci. 2023;35(1):2227780.
67. Hnamte V, Hussain J. Dependable intrusion detection system using deep convolutional neural network: a novel framework and performance evaluation approach. Telemat Inform Rep. 2023;11:100077.
68. Abdulganiyu OH, Tchakoucht TA, Saheed YK. Retracted article: towards an efficient model for network intrusion detection system (IDS): systematic literature review. Wirel Netw. 2024;30(1):453–82.
69. Satilmiş H, Akleylek S, Tok ZY. A systematic literature review on host-based intrusion detection systems. Ieee Access. 2024;12:27237–66.
70. Reddy DKK, Nayak J, Behera H, Shanmuganathan V, Viriyasitavat W, Dhiman G. A systematic literature review on swarm intelligence based intrusion detection system: past, present and future. Arch Comput Methods Eng. 2024. https://doi.org/10.1007/s11831-023-10059-2.
71. Fraley JB. The promise of machine learning in cybersecurity. In: SoutheastCon 2017. pp. 1–6. https://doi.org/10.1109/SECON.2017.7925283.
72. Dong B, Wang X. Comparison deep learning method to traditional methods using for network intrusion detection. In: 2016 8th IEEE Int. Conf. Commun. Softw. Networks, 2016. p. 581–585. https://doi.org/10.1109/ICCSN.2016.7586590.
73. Husain A, Salem A, Jim C, Dimitoglou G, College H. Development of an efficient network intrusion detection model using extreme gradient boosting (xgboost) on the unsw-nb15 dataset 2019.
74. Wang W, Sheng Y, Wang J, Zeng X, Ye X, Huang Y, et al. Hast-ids: learning hierarchical spatial-temporal features using deep neural networks to improve intrusion detection. IEEE Access. 2018;6:1792–806. https://doi.org/10.1109/ACCESS.2017.2780250.
75. Taher KA. Network intrusion detection using supervised machine learning technique with feature selection. In: 2019 Int. Conf. Robot. Signal Process. Tech., 2019. p. 643–646.
76. Hadri A. Intrusion detection system using pca and fuzzy pca techniques. In: 2016 Int. Conf. Adv. Commun. Syst. Inf. Secur., 2016. p. 1–7. https://doi.org/10.1109/ACOSIS.2016.7843930.
77. Jiang H, He Z, Ye G, Zhang H. Network intrusion detection based on pso-xgboost model. IEEE Access. 2020;8:58392–401. https://doi.org/10.1109/ACCESS.2020.2982418.
78. Bhamare D. Feasibility of supervised machine learning for cloud security. In: 2016 Int. Conf. Inf. Sci. Secur., 2016. p. 1–5. https://doi.org/10.1109/ICISSEC.2016.7885853.
79. Xie M, Hu J, Slay J. Evaluating host-based anomaly detection systems: Application of the one-class svm algorithm to. In: 2014 11th Int. Conf. Fuzzy Syst. Knowl. Discov., 2014. p. 978–982. https://doi.org/10.1109/FSKD.2014.6980972.
80. Chowdhury S, Khanzadeh M, Akula R, Zhang F, Zhang S, Medal H. Botnet detection using graph-based feature clustering. J Big Data. 2017. https://doi.org/10.1186/s40537-017-0074-7.
81. Shafiq M, Tian Z, Sun Y, Du X, Guizani M. Selection of effective machine learning algorithm and bot-iot attacks traffic identification for internet of things in smart city. Futur Gener Comput Syst. 2020;107:433–42. https://doi.org/10.1016/j.future.2020.02.017.
82. Ghosh P. Proposed ga-bfss and logistic regression based intrusion detection system. In: Proceedings of the 2015 Third International Conference on Computer Communication Control and Information Technology, 2015. p. 1–6. https://doi.org/10.1109/C3IT.2015.7060117.
83. Atienza D, Herrero, Corchado E. Neural analysis of http traffic for web attack detection. In: Proceedings of the 2015 International Conference on Data Mining and Applications, 2015. p. 201–212, https://doi.org/10.1007/978-3-319-19713-5.
84. Othman SM, Mutaher F, Alwi B, Alsohybe NT, Hashida AYA. Intrusion detection model using machine learning algorithm on big data environment. J Big Data. 2018. https://doi.org/10.1186/s40537-018-0145-4.
85. Firat I, Ertam F, Sengur A. Machine learning methods for cyber-security intrusion detection: datasets and comparative study. Comput Netw. 2021;188(January):107840. https://doi.org/10.1016/j.comnet.2021.107840.
86. Azam Z, Islam M, Huda MN. Comparative analysis of intrusion detection systems and machine learning-based model analysis through decision tree. IEEE Access. 2023;11:80348–91.
87. Iqbal A, Aftab S. A feed-forward and pattern recognition ann model for network intrusion detection. Int J Comput Netw Inf Secur. 2019. https://doi.org/10.5815/ijcnis.2019.04.03.
88. Dey P, Bhakta D. A new random forest and support vector machine-based intrusion detection model in networks. Natl Acad Sci Lett. 2023;46(5):471–7. https://doi.org/10.1007/s40009-023-01223-0.
89. Vishwakarma M, Kesswani N. A new two-phase intrusion detection system with naïve bayes machine learning for data classification and elliptic envelop method for anomaly detection. Decis Anal J. 2023;7:100233. https://doi.org/10.1016/j.dajour.2023.100233.
90. Alom Z, Taha TM. Network intrusion detection for cyber security using unsupervised deep learning approaches. 2017. p. 63–69.
91. Kodanda C, Srinivasa RT, Uma RE, Rao S. Attack classification in network intrusion detection system based on optimization strategy and deep learning methodology. Multimedia Tools Appl. 2024. https://doi.org/10.1007/s11042-024-18558-5.
92. Control J, Biju A, Franklin SW. Evaluated bird swarm optimization based on deep belief network (ebso-dbn) classification technique for iot network intrusion detection 2024, https://doi.org/10.1080/00051144.2023.2269646.
93. Bhati NS, Khari M, García-Díaz V, Verdú E. A review on intrusion detection systems and techniques. International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems. 2020;28(Supp02):65–91.
94. Bhati NS, Khari M. A new ensemble based approach for intrusion detection system using voting. J Intelligent Fuzzy Syst. 2022;42(2):969–79.

95. Ever E, Al-Turjman F. Forthcoming networks and sustainability in the iot era. Lecture Notes of the Institute for Computer Sciences. Soc Inform Telecommun Eng. 353
96. Kumar R, Quang NH, Solanki VK, Cardona M, Pattnaik PK. Research in Intelligent and Computing in Engineering: Select Proceedings of RICE 2020. Springer; 2021.
97. Bhati NS, Khari M. An ensemble model for network intrusion detection using adaboost, random forest and logistic regression. In: Applications of Artificial Intelligence and Machine Learning: Select Proceedings of ICAAAIML 2021. Springer; 2022. p. 777–789.
98. Bhati BS, Dikshita Bhati NS, Chugh G. A comprehensive study of intrusion detection and prevention systems. Wireless Commun Security, 2022;115–142.
99. Nagulapati VM, Raza Ur Rehman HM, Haider J, Abdul Qyyum M, Choi GS, Lim H. Hybrid machine learning-based model for solubilities prediction of various gases in deep eutectic solvent for rigorous process design of hydrogen purification. Sep Purif Technol. 2022;298:121651. https://doi.org/10.1016/j.seppur.2022.121651.
100. Chanakya G, Kunal P, Sumedh S, Priyanka W, Pn M. Network intrusion prevention system using machine learning techniques. 2017.
101. Kumar M. Intrusion detection system using decision tree algorithm. In: 2012 IEEE 14th International Conference on Communication Technology, 2012. p. 629–634. https://doi.org/10.1109/ICCT.2012.6511281.
102. Rathore MM, Saeed F, Rehman A, Paul A, Daniel A. Intrusion detection using decision tree model in high-speed environment. In: 2018 International Conference on Soft-computing Network Security, 2018. p. 1–4.
103. Rao BB, Swathi K. Fast knn classifiers for network intrusion detection system. Indian J Sci Technol. 2017;10:1–10. https://doi.org/10.17485/ijst/2017/v10i14/93690.
104. Shaukat K, Luo S, Varadharajan V, Member S. A survey on machine learning techniques for cyber security in the last decade. IEEE access. 2020;8:222310–54. https://doi.org/10.1109/ACCESS.2020.3041951.
105. Farnaaz N, Jabbar MA. Random forest modeling for network intrusion detection system. Procedia Comput Sci. 2016;89:213–7. https://doi.org/10.1016/j.procs.2016.06.047.
106. Sharma R, Kalita HK. Analysis of machine learning techniques based intrusion detection systems. 2016. https://doi.org/10.1007/978-81-322-2529-4.
107. Sen S, Gupta KD. Leveraging machine learning approach to setup software-defined network (sdn) controller rules during ddos attack. 2020. https://doi.org/10.1007/978-981-13-7564-4.
108. Rehman HMRU, On B-W, Ningombam DD, Yi S, Choi GS. Qsod: hybrid policy gradient for deep multi-agent reinforcement learning. IEEE Access. 2021;9:129728–41. https://doi.org/10.1109/ACCESS.2021.3113350.
109. Nandy T, Noor R, Kolandaisamy R, Yamani M, Idris I. A review of security attacks and intrusion detection in the vehicular networks. J King Saud Univ Comput Inform Sci. 2024;36(2):101945. https://doi.org/10.1016/j.jksuci.2024.101945.
110. Zhang Y, Li P, Wang X. Intrusion detection for IoT based on improved genetic algorithm and deep belief network. IEEE Access. 2019;7:31711–22. https://doi.org/10.1109/ACCESS.2019.2903723.
111. Sirisha A. A brief analysis on efficient machine learning techniques for intrusion detection model to provide network security. In: 2023 Int. Conf. Sustain. Comput. Data Commun. Syst., 2023. p. 105–112, https://doi.org/10.1109/ICSCDS56580.2023.10105087.
112. Siddiqui A, Elsalamouny M, Yoo PD, Muhaidat S, Kim K. Machine-learning-based feature selection techniques for large-scale network intrusion detection. In: 2014 IEEE 34th International Conference on Distributed Computing Systems Workshops, 2014. p. 177–181. https://doi.org/10.1109/ICDCSW.2014.14
113. Garg T, Student MT. Comparison of classification techniques for intrusion detection dataset using comparison of classification techniques for intrusion detection dataset using weka. 2018. https://doi.org/10.1109/ICRAIE.2014.6909184.
114. Relan NG, Student PG. Implementation of network intrusion detection system using variant of decision tree algorithm. In: 2015 Int. Conf. Nascent Technol. Eng. F., 2015. p. 1–5. https://doi.org/10.1109/ICNTE.2015.7029925.
115. Kotpalliwar MV. Classification of attacks using support vector machine (svm) on kddcup'99 ids database. In: 2015 Fifth Int. Conf. Commun. Syst. Netw. Technol., 2015. p. 987–990. https://doi.org/10.1109/CSNT.2015.185.
116. Subba B, Biswas S, Karmakar S. Enhancing performance of anomaly based intrusion detection systems through dimensionality reduction using principal component analysis. 2017. https://doi.org/10.1109/ANTS.2016.7947776.
117. Kevric J, Jukic S, Subasi A. An effective combining classifier approach using tree algorithms for network intrusion detection. Neural Comput Appl. 2017;28(s1):1051–8. https://doi.org/10.1007/s00521-016-2418-1.
118. Xu B, Chen S, Zhang H. Incremental k-nn svm method in intrusion detection.
119. Le T, Kim J, Kim H. An effective intrusion detection classifier using long short-term memory with gradient descent optimization. 2017. https://doi.org/10.1109/PlatCon.2017.7883684.
120. Zhang Q, Qu Y, Deng A. Network intrusion detection using kernel-based fuzzy-rough feature selection. In: 2018 IEEE Int. Conf. Fuzzy Syst., 2018. p. 1–6.
121. Al-hawawreh M, Moustafa N, Sitnikova E. Identification of malicious activities in industrial internet of things based on deep learning models. J Inf Secur Appl. 2018;41:1–11. https://doi.org/10.1016/j.jisa.2018.05.002.
122. Jamal A, Farrukh M, Khan A. A hybrid technique using binary particle swarm optimization and decision tree pruning for network intrusion detection. Cluster Comput. 2018;21(1):667–80. https://doi.org/10.1007/s10586-017-0971-8.
123. Qureshi A. RNN-ABC: a new swarm optimization based technique for anomaly detection. Computers. 2019. https://doi.org/10.3390/computers8030059.
124. Zhou Y, Cheng G, Jiang S, Dai M. Building an efficient intrusion detection system based on feature selection and ensemble classifier. Comput Networks. 2020. https://doi.org/10.1016/j.comnet.2020.107247.
125. Vinayakumar R, Alazab M, Kp S, Poornachandran P, Al-nemrat A, Venkatraman S. Deep learning approach for intelligent intrusion detection system. IEEE Access. 2019. https://doi.org/10.1109/ACCESS.2019.2895334.
126. Sarnovsky M, Paralic J. Ss symmetry learning and knowledge model. 2020. p. 1–14.
127. Kumar P, Kumar AA, Sahayakingsly C, Udayakumar A. Analysis of intrusion detection in cyber attacks using deep learning neural networks, 2021. p. 2565–2584.
128. Asif M, Abbas S, Khan MA, Fatima A, Adnan M, Lee S. Mapreduce based intelligent model for intrusion detection using machine learning technique. J King Saud Univ Comput Inf Sci. 2022;34(10):9723–31. https://doi.org/10.1016/j.jksuci.2021.12.008.

129. Jaradat AS, Barhoush MM, Easa RB. Network intrusion detection system: machine learning approach (May). Intell Dec Technol 2022. https://doi.org/10.11591/ijeecs.v25.i2.pp1151-1158.
130. Hossain A, Islam S. Ensuring network security with a robust intrusion detection system using. Array. 2023;19(May):100306. https://doi.org/10.1016/j.array.2023.100306.
131. Attou H, Guezzaz A, Benkirane S, Azrour M, Farhaoui Y. Cloud-based intrusion detection approach using machine learning techniques. Big Data Min Anal. 2023;6(3):311–20. https://doi.org/10.26599/BDMA.2022.9020038.
132. Saleh HM, Marouane H, Fakhfakh A. Stochastic gradient descent intrusions detection for wireless sensor network attack detection system using machine learning. IEEE Access. 2024;12:3825–36. https://doi.org/10.1109/ACCESS.2023.3349248.
133. Raza SN, Rehman H, Lee SG, Sang Choi G. Artificial intelligence based camera calibration. In: 2019 15th International Wireless Communications and Mobile Computing Conference (IWCMC), 2019. p. 1564–1569. https://doi.org/10.1109/IWCMC.2019.8766666.
134. Shafique R, Rustam F, Choi GS, Posa SK, Jurcut AD. In-vehicle networks security using transfer learning approach against ai-generated cyberattacks. In: 2024 35th Irish Signals and Systems Conference (ISSC), 2024. p. 1–6. https://doi.org/10.1109/ISSC61953.2024.10603221.
135. Ferrag MA, Maglaras L, Janicke H, Smith R. Deep learning techniques for cyber security intrusion detection: A detailed analysis. In: 6th International Symposium for ICS & SCADA Cyber Security Research 2019 2019. BCS Learning & Development
136. Wu Y, Zou B, Cao Y. Current status and challenges and future trends of deep learning-based intrusion detection models. J Imaging. 2024. https://doi.org/10.3390/jimaging10100254.
137. Ambekar NG, Thokchom S. Ul-vae: an unsupervised learning approach for zero-day malware detection using variational autoencoder. In: 2024 International Conference on Computational Intelligence and Network Systems (CINS), 2024. p. 1–7. IEEE
138. Mishra N, Pandya S. Internet of things applications, security challenges, attacks, intrusion detection, and future visions: a systematic review. IEEE Access. 2021;9:59353–77.
139. El Akhdar A, Baidada C, Kartit A, Hanine M, García CO, Lara RG, et al. Exploring the potential of microservices in internet of things: a systematic review of security and prospects. Sensors. 2024;24(20):6771.
140. Farooq S, Altaf A, Shoaib M, Iqbal F, Samee NA, Alohali MA, et al. Analyzing post-quantum cryptographic algorithms efficiency for transport security layer. Comput Electr Eng. 2025;125:110437.
141. Diro AA, Chilamkurti N. Distributed attack detection scheme using deep learning approach for internet of things. Future Gener Comput Syst. 2018;82:761–8.
142. Budhya SS, Lohia S, Deepak G, N, H.K. A survey on intrusion detection systems using various data mining techniques in big data (January). 2022.
143. Ashraf I, Park Y, Hur S, Kim SW, Alroobaea R, Zikria YB, et al. A survey on cyber security threats in iot-enabled maritime industry. IEEE Trans Intell Transp Syst. 2022;24(2):2677–90.
144. Ashraf I, Narra M, Umer M, Majeed R, Sadiq S, Javaid F, et al. A deep learning-based smart framework for cyber-physical and satellite system security threats detection. Electronics. 2022;11(4):667.
145. Alturki N, Aljrees T, Umer M, Ishaq A, Alsubai S, Saidani O, et al. An intelligent framework for cyber-physical satellite system and iot-aided aerial vehicle security threat detection. Sensors (Basel). 2023;23(16):7154.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.